

6 Averages of x^n and x^t

6.1 Introduction

The Dirichlet average of x^n is

$$R_n(b, z) = \int_E (u \cdot z)^n d\mu_b(u), \quad n \in \mathbb{N}, \quad (1)$$

where $b \in \mathbb{C}_{>}^k$. By performing the integration we shall express R_n as a homogeneous polynomial in the z_i with coefficients that are rational functions of the b_i . This polynomial has a meaning also when $b \notin \mathbb{C}_{>}^k$ and thereby provides an analytic continuation in b .

By averaging the Taylor series of an analytic function f , we found in Section 5.7 the series expansion

$$F(b, z) = \sum_{n=0}^{\infty} \frac{1}{n!} f^{(n)}(\lambda) R_n(b, z - \lambda). \quad (2)$$

Since R_n has a meaning whether or not $b \in \mathbb{C}_{>}^k$, we shall be able to use this series in Section 6.3 for analytic continuation of F in b . Thereafter $F(b, z)$ will be free of the requirement $b \in \mathbb{C}_{>}^k$ which was essential for its definition by an integral average.

Another fundamental result is the generating relation discussed in Section 6.6. The generating relation and the polynomial representation provide easy ways of relating R_n to the polynomials of Legendre, Gegenbauer, and Chebyshev, to Fibonacci numbers and Stirling numbers, and to complete and elementary symmetric functions. The generating rela-

tion leads in Section 6.8 to a representation of the R function by a single integral rather than a multiple integral, and this frees $R_i(b, z)$ from the requirement that the z_i lie in a half-plane which does not contain the origin. The representation by a single integral is the gateway to the topics of Chapters 8 and 9 because it allows any integral of the form

$$\int (A + Bx)^\alpha \dots (M + Nx)^\mu dx, \quad (3)$$

where $A, B, \alpha, \dots, M, N, \mu$ are constants, to be expressed as an R function. Finally, the generating relation is extended to a bilateral generating relation in Section 6.11 in order to deduce the integral formula for a product of two Gegenbauer polynomials. This will be a critical ingredient in the proof of the addition theorem for Gegenbauer polynomials in Section 7.3.

A third topic in this chapter is the transformation theory of R functions, comprising a binomial theorem and a linear transformation of R_n and (in the case of two variables and equal b parameters) two quadratic transformations of R_i . A primitive case of the first quadratic transformation has important consequences for Bessel functions. The second quadratic transformation establishes the link between Gegenbauer polynomials and the Jacobi polynomials discussed in Chapter 7. It also contains the Landen transformation of the first complete elliptic integral in a form which leads naturally to the computational algorithm of Gauss's arithmetic-geometric mean. A combination of the two quadratic transformations yields Borchardt's algorithm for computing logarithms, inverse circular functions, and inverse hyperbolic functions.

6.2 Representation as a polynomial

If $m = (m_1, \dots, m_k) \in \mathbb{N}^k$, we define $\|m\| = m_1 + \dots + m_k$.

REPRESENTATION 6.2-1 Let $n \in \mathbb{N}$, $(b, z) \in \mathbb{C}_{>}^k \times \mathbb{C}^k$, and $c = \sum_{i=1}^k b_i$. Define $R_n(b, z)$ by Definition 5.7-1. Then

$$R_n(b, z) = \frac{n!}{(c, n)} \sum \frac{(b_1, m_1) \dots (b_k, m_k)}{m_1! \dots m_k!} z_1^{m_1} \dots z_k^{m_k}, \quad (1)$$

where the summation extends over all $m \in \mathbb{N}^k$ such that $\|m\| = n$.

Proof By the multinomial theorem (2.3-9),

$$(u \cdot z)^n = n! \sum \frac{u_1^{m_1} \dots u_k^{m_k}}{m_1! \dots m_k!} z_1^{m_1} \dots z_k^{m_k},$$

where the summation extends over all $m \in \mathbb{N}^k$ with $\|m\| = n$. Integration with respect to μ_b using (4.4-8) proves (1). $\square$

If $n = 1$ and $w_i = b_i/c$, $i = 1, \dots, k$, we find from (1) that

$$R_1(b, z) = \sum_{i=1}^k w_i z_i. \quad (2)$$

If all the b_i are real and positive, this is a weighted arithmetic mean of the z_i . If $k = 2$, we can rewrite (1) in the form

$$R_n(\beta, \beta'; x, y) = \frac{n!}{(\beta + \beta', n)} \sum_{m=0}^n \frac{(\beta, m)(\beta', n-m)}{m!(n-m)!} x^m y^{n-m}. \quad (3)$$

For example,

$$R_2(\beta, \beta'; x, y) = \frac{\beta(\beta+1)x^2 + 2\beta\beta'xy + \beta'(\beta'+1)y^2}{(\beta + \beta')(\beta + \beta' + 1)}. \quad (4)$$

From (3) we find

$$(\beta + \beta', n)R_n(\beta, \beta'; x, 0) = (\beta, n)x^n, \quad (5)$$

and from (1), more generally,

$$(c_k, n)R_n(b_1, \dots, b_k; z_1, \dots, z_p, 0, \dots, 0) = (c_p, n)R_n(b_1, \dots, b_p; z_1, \dots, z_p), \quad (6)$$

where $1 \leq p \leq k$ and $c_r = \sum_{i=1}^r b_i$.

THEOREM 6.2-2 For every $n \in \mathbb{N}$ define $R_n(b, z)$ on $\mathbb{C}_{>}^k \times \mathbb{C}^k$ by Definition 5.7-1. Let $c = \sum_{i=1}^k b_i$. Then the product $(c, n)R_n(b, z)$ and the ratio $R_n(b, z)/\Gamma(c)$ have holomorphic continuations to $\mathbb{C}^{2k}$.

Proof If $(b, z) \in \mathbb{C}_{>}^k \times \mathbb{C}^k$, then by (1),

$$(c, n)R_n(b, z) = n! \sum \frac{(b_1, m_1) \cdots (b_k, m_k)}{m_1! \cdots m_k!} z_1^{m_1} \cdots z_k^{m_k}, \quad (7)$$

$$\frac{R_n(b, z)}{\Gamma(c)} = \frac{1}{\Gamma(c+n)} [(c, n)R_n(b, z)]. \quad (8)$$

Since the right side of (7) is a polynomial in b and z , it provides explicitly the holomorphic continuation of the left side to $\mathbb{C}^{2k}$. In (8) the reciprocal of $\Gamma(b_1 + \cdots + b_k + n)$ is an entire function of b by Corollary 3.6-2, and its product with the polynomial in brackets is an entire function of b and z , providing the holomorphic continuation of the left side. $\square$

It is the ratio $R_n(b, z)/\Gamma(c)$ which will be useful in the next section, but for the moment we note that $R_n(b, z)$ itself may henceforth be considered well-defined provided $c \neq 0, -1, -2, \dots, -n+1$, so that $(c, n) \neq 0$. Examples of well-defined R_n with c a negative integer equal to or less than $-n$ are provided by the elementary symmetric functions.

EXAMPLE 6.2-3 (*Complete and elementary symmetric functions.*) The complete symmetric function of degree n in k variables is the sum of all monomials of degree n ,

$$C_n(z) = \sum z_1^{m_1} \cdots z_k^{m_k}, \quad (9)$$

where the summation extends over all $m \in \mathbb{N}^k$ with $\|m\| = n$. For example the complete symmetric function of degree 2 in ξ, η, ζ is

$$C_2(\xi, \eta, \zeta) = \xi^2 + \eta^2 + \zeta^2 + \xi\eta + \xi\zeta + \eta\zeta. \quad (10)$$

If we choose every b parameter in (1) to be unity, we find

$$C_n(z) = \frac{(k, n)}{n!} R_n(1, \dots, 1; z_1, \dots, z_k), \quad n \in \mathbb{N}. \quad (11)$$

The constant $(k, n)/n!$ is the number of terms in the sum (9). It occurs in (11) because $R_n(b, z)$ reduces to ζ^n when every z_i is equal to ζ . For instance, $R_2(1, 1, 1; \xi, \eta, \zeta)$ equals the right side of (10) divided by $(3, 2)/2! = 6$. [See (6.6-8) for a generating relation for complete symmetric functions.]

The elementary symmetric function in k variables of degree $n \leq k$ is the sum of all monomials of degree n in which no variable occurs to a power higher than the first:

$$E_n(z) = \sum z_1^{m_1} \cdots z_k^{m_k}, \quad (12)$$

where the summation extends over all $m \in \{0, 1\}^k$ with $\|m\| = n$. (The restrictions on the m_i in (9) and (12) are like the restrictions on the occupation numbers in the quantum statistics of n bosons and n fermions, respectively.) For example, the elementary symmetric functions of ξ, η, ζ are

$$\begin{aligned} E_0 &= 1, & E_1(\xi, \eta, \zeta) &= \xi + \eta + \zeta, \\ E_2(\xi, \eta, \zeta) &= \xi\eta + \xi\zeta + \eta\zeta, & E_3(\xi, \eta, \zeta) &= \xi\eta\zeta. \end{aligned} \quad (13)$$

Note that

$$(t - \xi)(t - \eta)(t - \zeta) = t^3 - E_1 t^2 + E_2 t - E_3, \quad (14)$$

and in general the coefficients of a polynomial are the elementary symmetric functions of its zeros. Because of this connection with the theory of equations, elementary symmetric functions were studied by Newton,

Maclaurin, and others in the seventeenth and eighteenth centuries [see Hardy (1952, p. 52) and Beckenbach (1961, p. 11)].

Choose every b parameter in (1) to be -1 , and note that $(-1, m_i)/m_i!$ is 1 if $m_i = 0$, -1 if $m_i = 1$, and 0 if $m_i > 1$. Thus we find

$$E_n(z) = \binom{k}{n} R_n(-1, \dots, -1; z_1, \dots, z_k), \quad n = 0, 1, \dots, k. \quad (15)$$

The binomial coefficient is the number of terms in the sum (12). Because the b parameters are negative, E_n has no integral representation of the form (6.1-1).

THEOREM 6.2-4 Let p and k be integers such that $1 \leq p \leq k$. Let $n \in \mathbb{N}$, $z \in \mathbb{C}^k$, $b \in \mathbb{C}^p$, $c = \sum_{i=1}^p b_i$, and assume $(c, n) \neq 0$. Then

$$R_n(b_1, \dots, b_p, 0, \dots, 0; z_1, \dots, z_k) = R_n(b_1, \dots, b_p; z_1, \dots, z_p). \quad (16)$$

Proof In (7), $b_i = 0$ entails $(b_i, m_i) = 0$ if $m_i \neq 0$. The only nonzero terms in the sum are those with $m_{p+1} = \dots = m_k = 0$, and hence (16) holds with both sides multiplied by (c, n) . Divide by $(c, n) \neq 0$. $\square$

According to (16), vanishing b parameters can simply be omitted along with their corresponding variables, provided at least one of the b_i is nonzero. We consider next a limit in which all b parameters tend to zero together.

THEOREM 6.2-5 Let $n \in \mathbb{N}$, $z \in \mathbb{C}^k$, $c \in \mathbb{C}$, $w \in \mathbb{C}^k$, and define $cw = (cw_1, \dots, cw_k)$. Assume $\sum_{i=1}^k w_i = 1$ and $(c, n) \neq 0$. Then

$$\lim_{c \rightarrow 0} R_n(cw, z) = \sum_{i=1}^k w_i z_i^n, \quad (17)$$

$$\lim_{c \rightarrow \infty} R_n(cw, z) = \left(\sum_{i=1}^k w_i z_i \right)^n. \quad (18)$$

Proof For every $m \in \mathbb{N}^k$ with $\|m\| = n$, define

$$h_m(cw) = \frac{(cw_1, m_1) \cdots (cw_k, m_k)}{(c, n)}. \quad (19)$$

Then

$$R_n(cw, z) = n! \sum h_m(cw) \frac{z_1^{m_1} \cdots z_k^{m_k}}{m_1! \cdots m_k!}, \quad (20)$$

where the summation extends over all $m \in \mathbb{N}^k$ with $\|m\| = n$. As $c \rightarrow \infty$,

$$\frac{(cw_i, m_i)}{c^{m_i}} = w_i \left(w_i + \frac{1}{c} \right) \cdots \left(w_i + \frac{m_i - 1}{c} \right) \rightarrow w_i^{m_i}$$

and

$$h_m(cw) \rightarrow w_1^{m_1} \cdots w_k^{m_k}, \quad c \rightarrow \infty. \quad (21)$$

This proves (18) by the multinomial theorem (2.3-9).

If m_i is positive, the Appell symbol

$$(cw_i, m_i) = cw_i(cw_i + 1) \cdots (cw_i + m_i - 1)$$

tends to zero as $c \rightarrow 0$ because c is the first factor. The ratio $(cw_i, m_i)/(c, n)$ has a nonzero limit because the first factors in numerator and denominator cancel. If both m_i and m_j are positive, $i \neq j$, then $(cw_i, m_i)(cw_j, m_j)/(c, n) \rightarrow 0$ because there are two factors c in the numerator and only one in the denominator. Thus $h_m(cw) \rightarrow 0$ if two or more of the m_i are positive. If only one of the m_i is positive, it must be equal to n , and $(cw_i, n)/(c, n) \rightarrow w_i$. Thus

$$h_m(cw) \rightarrow \sum_{i=1}^k w_i \delta(m_i, n), \quad c \rightarrow 0, \quad (22)$$

where δ is the Kronecker delta. This proves (17). $\square$

Remark If the w_i are positive weights, the right side of (17) is the weighted average of n th powers while the right side of (18) is the n th power of the weighted average. If c , n , and the z_i also are positive, $[R_n(cw, z)]^{1/n}$ is well-defined and tends to the arithmetic mean as $c \rightarrow \infty$ and to the weighted mean of order n as $c \rightarrow 0$. The mean of order two, for example, is the root-mean-square. For any positive c , $[R_n(cw, z)]^{1/n}$ is a generalized mean value (see the Notes at the end of the chapter).

If the b parameters are all equal, say $b = (\beta, \beta, \dots, \beta)$, the conclusion of Theorem 6.2-2 that $R_n(b, z)/\Gamma(c)$ is entire in b and z can be strengthened.

THEOREM 6.2-6 Let $b = (\beta, \beta, \dots, \beta) \in \mathbb{C}^k$, $z \in \mathbb{C}^k$, and $n \in \mathbb{N}$. Then $R_n(b, z)/\Gamma(k\beta) = 0$ if $-\beta \in \mathbb{N}$, and $\Gamma(\beta)R_n(b, z)/\Gamma(k\beta)$ is entire in β and z .

Remark If $k = 2$, it follows by (3.7-2) that $R_n(\beta, \beta; x, y)/\Gamma(\beta + \frac{1}{2})$ is entire in β , x , y . Note that $R_n(\beta, \dots, \beta; z_1, \dots, z_k)$ is analytic at $\beta = 0, -1, -2, \dots$.

Proof Let $s - 1$ be the largest integer not exceeding $(n - 1)/k$, whence $s - 1 \leq (n - 1)/k < s$. In each term on the right side of (7) at least one of the summation indices, say m_i , must satisfy $m_i \geq s$, because $m_j \leq s - 1$ for all j would imply $\|m\| \leq k(s - 1) \leq n - 1$, contradicting $\|m\| = n$. If $(\beta, s) = 0$, then $(\beta, m_i) = 0$ and $(k\beta, n)R_n(b, z) = 0$. If $\beta = -s, -s - 1, \dots$, then

$k\beta + n$ is an integer which does not exceed $n - ks < 1$, and so $1/\Gamma(k\beta + n) = 0$. Hence,

$$\frac{R_n(b, z)}{\Gamma(k\beta)} = \frac{1}{\Gamma(k\beta + n)} [(k\beta, n)R_n(b, z)] = 0, \quad \beta = 0, -1, -2, \dots$$

The product $\Gamma(\beta)R_n(b, z)/\Gamma(k\beta)$ is holomorphic in β and z except for the simple poles of $\Gamma(\beta)$ at $\beta = 0, -1, -2, \dots$. Let $p \in \mathbb{N}$. Since $R_n(b, z)/\Gamma(k\beta)$ is entire in β and z by Theorem 6.2-2 and vanishes if $\beta = -p$, it can be represented as $(\beta + p)g(\beta, z)$, where g is an entire function of β and z . But $\Gamma(\beta)(\beta + p)g(\beta, z)$ approaches a finite limit as $\beta \rightarrow -p$ by (3.3-3), and hence the singularity of $\Gamma(\beta)R_n(b, z)/\Gamma(k\beta)$ at $\beta = -p$ is removable. $\square$

Some inequalities for the absolute value of R_n will be needed in the next section. If $z = (z_1, \dots, z_k) \in \mathbb{C}^k$, we define two norms,

$$|z| = \max\{|z_1|, \dots, |z_k|\} \quad \text{and} \quad \|z\| = |z_1| + \dots + |z_k|. \quad (23)$$

The first is more frequently useful, but both occur in the present context.

INEQUALITIES 6.2-7 Let $n \in \mathbb{N}$, $z \in \mathbb{C}^k$, $b \in \mathbb{C}^k$, and $c = \sum_{i=1}^k b_i$. Then

$$|(c, n)R_n(b, z)| \leq (\|b\|, n)|z|^n, \quad (24)$$

$$|(c, n)R_n(b, z)| \leq (|b|, n)\|z\|^n. \quad (25)$$

Proof By (7) and (2.2-10),

$$|(c, n)R_n(b, z)| \leq n! \sum \frac{(|b_1|, m_1) \cdots (|b_k|, m_k)}{m_1! \cdots m_k!} |z_1|^{m_1} \cdots |z_k|^{m_k}. \quad (26)$$

The summation is restricted by the condition $\|m\| = n$, which implies

$$|z_1|^{m_1} \cdots |z_k|^{m_k} \leq |z|^n.$$

We replace one side of this last inequality by the other in (26), and perform the summation by Vandermonde's theorem (Theorem 2.3-2) to prove (24). On the other hand,

$$(|b_1|, m_1) \cdots (|b_k|, m_k) \leq (|b|, m_1) \cdots (|b|, m_k) \leq (|b|, n),$$

since the middle member is a product of n positive factors which do not exceed the n factors of $(|b|, n)$. Another replacement is made in (26), and the summation is accomplished by the multinomial theorem (2.3-9) to prove (25). $\square$

From (24) we now deduce an inequality which will be used in the next section to majorize a series of R polynomials and thereby prove its uniform convergence.

INEQUALITY 6.2-8 Let $n \in \mathbb{N}$, $r \in \mathbb{R}$, $z \in \mathbb{C}^k$, $b \in \mathbb{C}^k$, and $c = \sum_{i=1}^k b_i$. If $\|b\| \leq r \leq n-2$, then

$$\left| \frac{R_n(b, z)}{\Gamma(c)} \right| \leq e^{\pi r/2} \frac{(r, n)|z|^n}{\Gamma(n-r)}. \quad (27)$$

Proof By (24),

$$\left| \frac{R_n(b, z)}{\Gamma(c)} \right| = \left| \frac{(c, n)R_n(b, z)}{\Gamma(c+n)} \right| \leq \frac{(\|b\|, n)|z|^n}{|\Gamma(c+n)|} \leq \frac{(r, n)|z|^n}{|\Gamma(c+n)|}. \quad (28)$$

Since $|c| \leq \|b\| \leq r$ it follows that $|\operatorname{Im}(c+n)| = |\operatorname{Im} c| \leq r$ and $\operatorname{Re} c + n \geq n-r \geq 2$. By (3.10-7),

$$|\Gamma(c+n)| \geq \Gamma(\operatorname{Re} c + n)e^{-\pi r/2}. \quad (29)$$

Moreover, by (3.10-2) and Inequality 3.10-1, $\Gamma(\operatorname{Re} c + n) \geq \Gamma(n-r)$, and hence

$$|\Gamma(c+n)| \geq \Gamma(n-r)e^{-\pi r/2}. \quad (30)$$

By combining (28) and (30), (27) is proved. $\square$

6.3 Dirichlet averages with negative parameters

The ratio $R_n(b, z)/\Gamma(c)$ is an entire function of b according to Theorem 6.2-2. If $z_1, \dots, z_k$ are restricted to a disk on which f is holomorphic, we shall prove now that $F(b, z)/\Gamma(c)$ is an entire function of b and can be represented by an infinite series. (Only when $b \in \mathbb{C}_{>}^k$ is $F(b, z)/\Gamma(c)$ represented by the multiple integral.) In Theorem 6.3-6 a contour integral will allow the disk to be replaced by any convex set.

THEOREM 6.3-1 Let Ω be an open circular disk in $\mathbb{C}$ with center λ , and let f be holomorphic on Ω . Let $F(b, z)$ be defined on $\mathbb{C}_{>}^k \times \Omega^k$ by Definition 5.2-1, and let $c = \sum_{i=1}^k b_i$. Then $F(b, z)/\Gamma(c)$ has a holomorphic continuation to $\mathbb{C}^k \times \Omega^k$, where it is represented by

$$\frac{F(b, z)}{\Gamma(c)} = \sum_{n=0}^{\infty} \frac{f^{(n)}(\lambda)}{n!} \frac{R_n(b, z-\lambda)}{\Gamma(c)}. \quad (1)$$

Proof If $(b, z) \in \mathbb{C}_{>}^k \times \Omega^k$, then (1) is the series (5.7-4) divided by $\Gamma(c)$. Each term of the series is an entire function of b and z by Theorem 6.2-2. If we can show that the series converges uniformly on compact subsets of $\mathbb{C}^k \times \Omega^k$, it will follow [see Rudin (1974, Section 10.28)] that the sum of the series is holomorphic on $\mathbb{C}^k \times \Omega^k$ and thus provides the continuation of $F(b, z)/\Gamma(c)$.

Let ρ denote the radius of Ω . It suffices to prove uniform convergence on the compact subset

$$K(r, s) = \{(b, z) \in \mathbb{C}^{2k} : \|b\| \leq r, |z - \lambda| \leq s\}$$

for every $r > 0$ and every $s < \rho$. Let $u_n(b, z)$ denote the general term of (1), so that

$$|u_n(b, z)| = a_n \left| \frac{R_n(b, z - \lambda)}{\Gamma(c)} \right|, \quad a_n = \frac{|f^{(n)}(\lambda)|}{n!}.$$

If we assume $(b, z) \in K(r, s)$ and $n \geq r + 2$, then (6.2-27) implies

$$|u_n(b, z)| \leq a_n e^{\pi r/2} \frac{(r, n)s^n}{\Gamma(n - r)}.$$

Therefore, if we omit terms with $n < r + 2$, the series (1) is majorized by

$$\frac{e^{\pi r/2}}{\Gamma(r)} \sum \frac{\Gamma(n + r)}{\Gamma(n - r)} a_n s^n.$$

The root test for convergence is unaffected by the ratio of gamma functions, the n th root of which tends to unity by Exercise 3.4-5. Moreover, $\rho \limsup a_n^{1/n} \leq 1$ because the Taylor series of f about λ converges absolutely on Ω . Since $s < \rho$, the majorizing series converges. By the Weierstrass M test the series (1) converges uniformly on $K(r, s)$. $\square$

We may cancel $\Gamma(c)$ on the two sides of (1) if $c \neq 0, -1, -2, \dots$. Thus, it is convenient to define

$$U_k = \left\{ b \in \mathbb{C}^k : \sum_{i=1}^k b_i \neq 0, -1, -2, \dots \right\}. \quad (2)$$

Note that $U_1 = U$ (see Definition 3.2-2) and $b \in U_k$ implies $c \in U$. Do not confuse U_k with the Cartesian power U^k : neither set contains the other.

COROLLARY 6.3-2 Assume $1 \leq p \leq k$, $b = (b_1, \dots, b_p, 0, \dots, 0) \in U_k$, and $z \in \Omega^k$. Then

$$F(b_1, \dots, b_p, 0, \dots, 0; z_1, \dots, z_k) = F(b_1, \dots, b_p; z_1, \dots, z_p). \quad (3)$$

Proof Equation (1) can be multiplied by $\Gamma(c)$ since $b \in U_k$. Vanishing b parameters can be omitted along with their corresponding variables in R_n by Theorem 6.2-4. Hence the same is true of F . $\square$

COROLLARY 6.3-3 Let $S(b, z)$ be defined on $\mathbb{C}_{>}^k \times \mathbb{C}^k$ by Definition 5.8-1. Then $S(b, z)/\Gamma(c)$ has a holomorphic continuation to $\mathbb{C}^{2k}$.

Proof If $z \in \mathbb{C}^k$, there exists an open circular disk $\Omega \subset \mathbb{C}$ such that $z \in \Omega^k$, and it suffices to prove the existence of a continuation to $\mathbb{C}^k \times \Omega^k$. The existence was proved by Theorem 6.3-1. $\square$

COROLLARY 6.3-4 Let H be an open half-plane in $\mathbb{C} - \{0\}$, and let $R_t(b, z)$ be defined by Definition 5.9-1 for every $(t, b, z) \in \mathbb{C} \times \mathbb{C}_{>}^k \times H^k$. Then $R_t(b, z)/\Gamma(c)$ has a holomorphic continuation to $\mathbb{C}^{k+1} \times H^k$.

Proof If $z \in H^k$, there exists an open circular disk $\Omega \subset H$ such that $z \in \Omega^k$, and it suffices to prove the existence of a continuation to $\mathbb{C}^{k+1} \times \Omega^k$. For every fixed $t \in \mathbb{C}$, $R_t(b, z)/\Gamma(c)$ has a holomorphic continuation in (b, z) to $\mathbb{C}^k \times \Omega^k$ by Theorem 6.3-1. Since a function is holomorphic in several variables jointly if it is holomorphic in each one separately, we need only show that $R_t(b, z)/\Gamma(c)$ is entire in t for every fixed $(b, z) \in \mathbb{C}^k \times \Omega^k$.

Denote the center of Ω by λ and put $f(x) = x^t$ in (1). Then

$$\frac{R_t(b, z)}{\Gamma(c)} = \lambda^t \sum_{n=0}^{\infty} \frac{(-1)^n (-t, n)}{n!} \lambda^{-n} \frac{R_n(b, z - \lambda)}{\Gamma(c)}.$$

Since $\lambda \neq 0$, the factor λ^t is entire in t . Given $T > 0$, the series is majorized for $|t| \leq T$ by $\sum u_n$, where

$$u_n = \frac{(T, n)}{n!} |\lambda|^{-n} \left| \frac{R_n(b, z - \lambda)}{\Gamma(c)} \right|.$$

By Exercises 3.4-5 and 6.2-7,

$$\limsup_{n \rightarrow \infty} u_n^{1/n} \leq \left| \frac{z - \lambda}{\lambda} \right|.$$

Since $z \in \Omega^k$ implies $|z - \lambda| < |\lambda|$, $\sum u_n$ converges by the root test, and the original series converges uniformly for $|t| \leq T$ by the Weierstrass M test. Because each term is a polynomial in t , the sum of the series is holomorphic for $|t| \leq T$, where T is arbitrarily large. $\square$

COROLLARY 6.3-5 Let $n \in \mathbb{N}$, $b \in \mathbb{C}^k$, $z \in \mathbb{C}^k$, $s \in \mathbb{C}$, $s - z = (s - z_1, \dots, s - z_k)$, and $c = \sum_{i=1}^k b_i$. Let $\text{con}(z)$ denote the convex hull of $\{z_1, \dots, z_k\}$, and define $\mathbb{C}' = \mathbb{C} - \text{con}(z)$. Then $R_{-n-1}(b, s - z)/\Gamma(c)$ is holomorphic in (b, s) on $\mathbb{C}^k \times \mathbb{C}'$.

Proof The set $\mathbb{C}'$ is a union of open half-planes not containing $z_1, \dots, z_k$. If J is one of these half-planes, $s \in J$ implies that $s - z_1, \dots, s - z_k$ lie in a half-plane H not containing 0. By Corollary 6.3-4, $R_t(b, s - z)$ is holomorphic in $(b, s - z)$ on $\mathbb{C}^k \times H^k$ and therefore in (b, s) on $\mathbb{C}^k \times J$. Thus $R_{-n-1}(b, s - z)$ is holomorphic in (b, s) on $\mathbb{C}^k \times \mathbb{C}'$ if it returns to its initial value when s encircles $\text{con}(z)$, say in the positive

direction. Since every component of $s - z$ returns to its initial value multiplied by $e^{i2\pi}$, and since R_{-n-1} is homogeneous of degree $-n-1$, the value of the function is multiplied by $(e^{i2\pi})^{-n-1} = 1$. $\square$

THEOREM 6.3-6 Let Ω be a convex open set in $\mathbb{C}$, and let f be holomorphic on Ω . Let $F^{(n)}(b, z)$ be defined on $\mathbb{C}_{>}^k \times \Omega^k$ for every $n \in \mathbb{N}$ by Definition 5.3-1, and let $c = \sum_{i=1}^k b_i$. Then $F^{(n)}(b, z)/\Gamma(c)$ has a holomorphic continuation to $\mathbb{C}^k \times \Omega^k$, where it is represented as follows: If γ is a positively oriented rectifiable Jordan curve in Ω such that $\text{con}(z) \subset I(\gamma)$, then

$$\frac{F^{(n)}(b, z)}{\Gamma(c)} = \frac{n!}{2\pi i} \int_{\gamma} f(s) \frac{R_{-n-1}(b, s - z)}{\Gamma(c)} ds. \quad (4)$$

Remark Note that the circular disk of Theorem 6.3-1 has been replaced by a convex set. Although the convex set is a union of circular disks, the change is not trivial because the Cartesian power of a union of sets is not in general contained in the union of the Cartesian powers.

Proof If $b \in \mathbb{C}^k$ and $z \in \Omega^k$, the integrand of (4) is holomorphic in s on $\Omega - \text{con}(z)$ by Corollary 6.3-5. By Cauchy's integral theorem the value of the integral is the same for every closed contour γ which lies in Ω and encircles $\text{con}(z)$ once in the positive direction. Hence there is no loss of generality in choosing γ to be the positively oriented boundary of a convex open bounded set D such that $\text{con}(z) \subset D$ and the boundary of D is in Ω . (For example we can choose D to be an ε neighborhood of $\text{con}(z)$ for sufficiently small ε .)

If D is fixed while b and z vary, (4) holds on $\mathbb{C}_{>}^k \times D^k$ by (5.11-2). We need only show that the integral is holomorphic on $\mathbb{C}^k \times D^k$, because it will then provide explicitly the continuation of the left side. The existence of a holomorphic continuation to $\mathbb{C}^k \times \Omega^k$ will follow, since Ω^k is simply connected and every point in Ω^k has a neighborhood like D^k .

To prove that the integral is holomorphic on $\mathbb{C}^k \times D^k$, we denote the integrand by $\varphi(s, b, z)$ and verify the three hypotheses of Theorem B.4.

(i) For each fixed $s \in \gamma$, there exists (by the convexity of D) an open half-plane H such that $D \subset H$ and $s \notin H$. By Corollary 6.3-4 (with a shift of origin) it follows that $[(b, z) \mapsto \varphi(s, b, z)]$ is holomorphic on $\mathbb{C}^k \times H^k$ and therefore on $\mathbb{C}^k \times D^k$.

(ii) For each fixed $(b, z) \in \mathbb{C}^k \times D^k$, $[s \mapsto \varphi(s, b, z)]$ is holomorphic on $\Omega - \text{con}(z)$ by Corollary 6.3-5 and therefore integrable on γ .

(iii) φ is continuous on $\gamma \times \mathbb{C}^k \times D^k$ because of the analyticity properties stated in (i) and (ii). If K is a compact set in $\mathbb{C}^k \times D^k$, then φ is bounded on the compact set $\gamma \times K$, and we can choose g_K to be a constant.

It follows from Theorem B.4 that the integral is holomorphic on $\mathbb{C}^k \times D^k$. $\square$

Various relations for F , S , and R were established in Chapter 5 subject to the restriction $b \in \mathbb{C}_{>}^k$. By the permanence of functional relations [see Ahlfors (1966, Chapter 8, Section 1.1) or Hille (1959, Vol. II, Section 10.7)], which asserts that a functional relation between holomorphic functions holds also between their holomorphic continuations, this restriction can now be dropped after dividing by a suitable gamma function. Instead of dividing by a gamma function, it is sometimes more convenient to exclude its poles. For example it follows from case (ii) of Theorem 5.4-1 that $F(b, z)/\Gamma(c)$ is a solution of the Euler–Poisson equations on Ω^k for any complex b . If $b \in U_k$ we may multiply by $\Gamma(c)$ to prove that $F(b, z)$ is a solution. The Remark at the end of Section 5.6 and the Remark following Proposition 5.9-8 have similar justifications. In the definition (5.5-14) of the Riemann–Liouville fractional integral, $(z - a)^\nu$ is an entire function of ν and so is $F(1, \nu; z, a)/\Gamma(1 + \nu)$. Finally, if the series (5.8-4) and (5.9-4) are divided by $\Gamma(c)$ to make them special cases of (1), they become valid for all complex values of the parameters:

$$\frac{S(b, z)}{\Gamma(c)} = \sum_{n=0}^{\infty} \frac{1}{n!} \frac{R_n(b, z)}{\Gamma(c)}, \quad z \in \mathbb{C}^k, \quad (5)$$

$$\frac{R_{-a}(b, z)}{\Gamma(c)} = \sum_{n=0}^{\infty} \frac{(a, n)}{n!} \frac{R_n(b, 1 - z)}{\Gamma(c)}, \quad |1 - z| < 1. \quad (6)$$

Dirichlet averages with equal b parameters are important because they include the functions of Bessel, Legendre, and Gegenbauer. Although Theorem 6.3-1 leads us to expect possible singularities of $F(\beta, \beta, \dots, \beta; z_1, \dots, z_k)$ when $k\beta = 0, -1, -2, \dots$, we shall see now that the ones with $\beta = 0, -1, -2, \dots$ are removable.

COROLLARY 6.3-7 Let $b = (\beta, \beta, \dots, \beta) \in \mathbb{C}^k$. Let Ω be a convex open set in $\mathbb{C}$ and let f be holomorphic on Ω . Then $\Gamma(\beta)F(b, z)/\Gamma(k\beta)$ is holomorphic in (β, z) on $\mathbb{C} \times \Omega^k$.

Remark If $k = 2$, it follows by (3.7-2) that $F(\beta, \beta; x, y)/\Gamma(\beta + \frac{1}{2})$ is holomorphic in (β, x, y) on $\mathbb{C} \times \Omega^2$. In particular the Bessel function defined by (5.8-23) has the property that $y^{-\mu}J_\mu(y)$ is entire in μ and y .

Proof If every component of z lies in an open circular disk in Ω , then $F(b, z)/\Gamma(k\beta)$ is represented by (1) and vanishes if $-\beta \in \mathbb{N}$ as a result of Theorem 6.2-6. By the permanence of functional relations $F(b, z)/\Gamma(k\beta) = 0$ for every $z \in \Omega^k$ if $-\beta \in \mathbb{N}$.

By Theorems 6.3-6 and 3.3-1, $\Gamma(\beta)F(b, z)/\Gamma(k\beta)$ is holomorphic in (β, z) on $U \times \Omega^k$. If $n \in \mathbb{N}$, $\Gamma(\beta)$ has a simple pole at $\beta = -n$, while $F(b, z)/\Gamma(k\beta)$ is entire in β and has a zero at $\beta = -n$. Hence their product tends to a finite limit as $\beta \rightarrow -n$, and the singularity is removable. $\square$

6.4 The binomial theorem

The ordinary binomial theorem (2.3-8) for x^n ,

$$(x + \lambda)^n = \sum_{m=0}^n \binom{n}{m} \lambda^{n-m} x^m, \quad n \in \mathbb{N}, \quad (1)$$

implies a corresponding theorem for the Dirichlet average of x^n .

THEOREM 6.4-1 Let $n \in \mathbb{N}$, $b \in \mathbb{C}^k$, $z \in \mathbb{C}^k$, and $\lambda \in \mathbb{C}$. Define $c = \sum_{i=1}^k b_i$ and assume $(c, n) \neq 0$. Then

$$R_n(b, z + \lambda) = \sum_{m=0}^n \binom{n}{m} \lambda^{n-m} R_m(b, z), \quad (2)$$

where $z + \lambda = (z_1 + \lambda, \dots, z_k + \lambda)$.

Proof The case $k = 1$ is (1). If $k \geq 2$ and $b \in \mathbb{C}_{>}^k$, Definition 5.7-1 shows that

$$\begin{aligned} R_n(b, z + \lambda) &= \int_E [u \cdot (z + \lambda)]^n d\mu_b(u) = \int_E (u \cdot z + \lambda)^n d\mu_b(u) \\ &= \sum_{m=0}^n \binom{n}{m} \lambda^{n-m} \int_E (u \cdot z)^m d\mu_b(u), \end{aligned}$$

which is the same as (2). By Theorem 6.2-2 and the permanence of functional relations, (2) is valid for every $b \in \mathbb{C}^k$ such that $(c, n) \neq 0$. $\square$

It is easy to show that (2) is equivalent to a differential-difference equation for R_n . More generally, a sequence of polynomials $\{p_0(z), p_1(z), p_2(z), \dots\}$ in k variables ($z \in \mathbb{C}^k$) satisfies a binomial theorem,

$$p_n(z + \lambda) = \sum_{m=0}^n \binom{n}{m} \lambda^{n-m} p_m(z), \quad n \in \mathbb{N}, \quad (3)$$

if and only if

$$\sum_{i=1}^k D_i p_n = n p_{n-1}, \quad n \in \mathbb{N}, \quad (4)$$

where the right side is taken to be zero if $n = 0$ [see Carlson (1970b)]. In

the case $k = 1$ such sets of polynomials were first discussed by Appell (1880b), and many special polynomials in one variable such as those of Bernoulli, Euler, Legendre, Gegenbauer, Hermite, and Laguerre form Appell sets. Many other special functions which are not polynomials satisfy

$$\frac{df_t}{dz} = tf_{t-1}, \quad (5)$$

where t need not be an integer. The solutions of this equation [strictly speaking, the equation satisfied by $g_a(z) = f_{-a}(z)/\Gamma(-a)$] have been studied by Truesdell (1948). An extension of (5) to several variables is

$$\sum_{i=1}^k D_i f_t = tf_{t-1}, \quad (6)$$

and one solution is $R_i(b, z)$ according to (5.9-2).

6.5 Linear transformation

An equality between two R polynomials will be called a linear transformation if the arguments of one polynomial are linear functions of the arguments of the other. We first prove a linear transformation for the case $k = 2$. One of the arguments remains unchanged, and the other is replaced by their difference.

TRANSFORMATION 6.5-1 Let $n \in \mathbb{N}$ and $\beta, \beta', x, y \in \mathbb{C}$. Define $c = \beta + \beta'$. Then

$$(c, n)R_n(\beta, \beta'; x, y) = (\beta, n)R_n(1 - c - n, \beta'; x, x - y) \quad (1)$$

$$= (\beta', n)R_n(\beta, 1 - c - n; y - x, y). \quad (2)$$

Proof Because of the symmetry property of R_n , it is sufficient to prove (2), since (1) and (2) differ merely by interchange of β with β' and x with y . Using the binomial theorem (6.4-2) and (6.2-5), we find

$$\begin{aligned} (c, n)R_n(\beta, \beta'; x, y) &= (c, n) \sum_{m=0}^n \binom{n}{m} y^{n-m} R_m(\beta, \beta'; x - y, 0) \\ &= \sum_{m=0}^n \binom{n}{m} y^{n-m} \frac{(c, n)(\beta, m)}{(c, m)} (x - y)^m. \end{aligned}$$

By (2.2-9),

$$(c, n)/(c, m) = (-1)^{n-m}(1 - c - n, n - m).$$

Hence

$$\begin{aligned}(c, n)R_n(\beta, \beta'; x, y) &= n!(-1)^n \sum_{m=0}^n \frac{(\beta, m)(1-c-n, n-m)}{m!(n-m)!} (y-x)^m y^{n-m} \\ &= (-1)^n (\beta + 1 - c - n, n)R_n(\beta, 1 - c - n; y - x, y).\end{aligned}$$

Finally, by (2.2-5) and (2.2-6),

$$(\beta + 1 - c - n, n) = (1 - \beta' - n, n) = \frac{1}{(1 - \beta', -n)} = (-1)^n (\beta', n). \quad \square$$

By combining the linear transformation with the symmetry property, we can express an R polynomial with arguments (x, y) as an R polynomial of the same degree with arguments (ξ, η) , for which there are six possibilities: (x, y) , (y, x) , $(x, x - y)$, $(x - y, x)$, $(y - x, y)$, $(y, y - x)$. By Proposition 5.9-7 an R polynomial with arguments (x, y) or (ξ, η) is equivalent to a ${}_2F_1$ polynomial with argument $z = 1 - x/y$ or $\zeta = 1 - \xi/\eta$, respectively. Hence a ${}_2F_1$ polynomial with argument z can be expressed as a ${}_2F_1$ polynomial with argument ζ , for which there are six possibilities: z , $z/(z - 1)$, $1/z$, $1/(1 - z)$, $1 - z$, $(z - 1)/z$. For example, $(\xi, \eta) = (x, x - y)$ implies

$$\zeta = 1 - \xi/\eta = 1 - x/(x - y) = -y/(x - y) = (1 - x/y)^{-1} = 1/z.$$

Thus there are six ways of writing any ${}_2F_1$ polynomial, and the transformations connecting them, which are easily deduced from (2) and (5.9-11) (see Exercise 6.5-5) form a group of order six. The transformation corresponding to $\zeta = z/(z - 1)$, which also holds for nonterminating ${}_2F_1$ series, is Pfaff's transformation (5.9-14).

The linear transformation of R_n is easily generalized to any number of variables (see Exercise 6.5-1):

$$\begin{aligned}(c, n)R_n(b_1, \dots, b_k; z_1, \dots, z_k) \\ = (b_1, n)R_n(1 - c - n, b_2, \dots, b_k; z_1, z_1 - z_2, \dots, z_1 - z_k).\end{aligned} \quad (3)$$

Because of the symmetry of R_n , (3) is one of k equations which differ by permutations of the indices. For example, if $k = 2$ there are two equations, (1) and (2).

6.6 Generating functions

The R polynomials have a simple generating function which includes as special cases the generating functions of Legendre and Gegenbauer polynomials, complete and elementary symmetric functions, and Fibonacci and Stirling numbers. The generating relation will be used in Section 6.8 to deduce an important integral representation of the R function.

GENERATING RELATION 6.6-1 Let $b \in \mathbb{C}^k$, $z \in \mathbb{C}^k$, $t \in \mathbb{C}$, and $c = \sum_{i=1}^k b_i$. Assume $|tz| < 1$, where $tz = (tz_1, \dots, tz_k)$ and the norm is defined by (6.2-23). Then

$$\prod_{i=1}^k (1 - tz_i)^{-b_i} = \sum_{n=0}^{\infty} t^n \frac{(c, n)}{n!} R_n(b, z). \quad (1)$$

Remark Since R_n is already known to be homogeneous of degree n in z , there is no loss of generality in setting $t = 1$. However, if R_n had not been previously defined, (1) with t included could be used as a definition. Special cases such as the Legendre polynomials are often defined in this way.

Proof For each $i = 1, \dots, k$, the binomial series

$$(1 - tz_i)^{-b_i} = \sum_{m_i=0}^{\infty} t^{m_i} \frac{(b_i, m_i)}{m_i!} z_i^{m_i}$$

is absolutely convergent since $|tz_i| \leq |tz| < 1$. If $m = (m_1, \dots, m_k)$ and $\|m\| = m_1 + \dots + m_k$, the product of the k series [see Apostol (1974, p. 203)] is

$$\prod_{i=1}^k (1 - tz_i)^{-b_i} = \sum_{m_1=0}^{\infty} \dots \sum_{m_k=0}^{\infty} t^{\|m\|} \frac{(b_1, m_1) \dots (b_k, m_k)}{m_1! \dots m_k!} z_1^{m_1} \dots z_k^{m_k}.$$

The multiple series converges absolutely and can be rearranged in any order. By (6.2-1) it is the same as the right side of (1). $\square$

THEOREM 6.6-2 Let $n \in \mathbb{N}$, $z \in \mathbb{C}^k$, and $b \in U^k \cap U_k$. Then

$$\limsup_{n \rightarrow \infty} |R_n(b, z)|^{1/n} = |z|. \quad (2)$$

Proof Since $b \in U^k$, the left side of (1) has a singularity in the t plane at z_i^{-1} if $z_i \neq 0$. The circle of convergence of the Taylor series passes through the singularity closest to the origin and has radius $|z|^{-1}$. From the root test it follows that

$$|z| = \limsup_{n \rightarrow \infty} \left| \frac{(c, n)}{n!} R_n(b, z) \right|^{1/n}.$$

Since $b \in U_k$ implies $c \in U$, the n th root of $|(c, n)/n!|$ tends to 1 by Exercise 3.4-5. $\square$

More complicated generating functions are furnished by the series expansion (6.3-1), which can be rewritten as

$$F(b, \lambda + tz) = \sum_{n=0}^{\infty} t^n \frac{f^{(n)}(\lambda)}{n!} R_n(b, z), \quad tz \in \Omega^k, \quad (3)$$

where $b \in U_k$ and Ω is an open circular disk in the ζ plane with center 0 on which $f(\lambda + \zeta)$ is holomorphic in ζ . In particular, if Ω is the unit disk, $\lambda = 1$, and $f(1 + \zeta) = (1 + \zeta)^{-a}$, we recover (6.3-6) in the form

$$R_{-a}(b, 1 - tz) = \sum_{n=0}^{\infty} t^n \frac{(a, n)}{n!} R_n(b, z), \quad |tz| < 1. \quad (4)$$

Comparison with (1) shows that

$$R_{-c}(b, z) = \prod_{i=1}^k z_i^{-b_i}, \quad b \in U_k, \quad c = \sum_{i=1}^k b_i, \quad (5)$$

where $|1 - z| < 1$. This restriction can be relaxed because both sides of (5) have holomorphic continuations to H^k if H is an open half-plane such that $1 \in H$ and $0 \notin H$. By the permanence of functional relations, (5) is true for every $z \in H^k$. We have already encountered the case $k = 2$ in (5.9-22).

If $z \in H^k$ and $b \in \mathbb{C}_{>}^k$, $k \geq 2$, we conclude from (5) and (5.2-1) that

$$\prod_{i=1}^k z_i^{-b_i} = \int_E (u \cdot z)^{-c} d\mu_b(u). \quad (6)$$

Special cases in which the b parameters are positive integers, such as

$$\frac{1}{xy} = \int_0^1 [ux + (1-u)y]^{-2} du, \quad (7)$$

are used to evaluate certain integrals in quantum electrodynamics by a technique due to Feynman (1949).

For the complete symmetric functions $C_n(z)$ in (6.2-11), the generating relation (1) becomes

$$\prod_{i=1}^k (1 - tz_i)^{-1} = \sum_{n=0}^{\infty} t^n C_n(z), \quad |tz| < 1. \quad (8)$$

For the elementary symmetric functions $E_n(z)$ in (6.2-15),

$$\prod_{i=1}^k (1 - tz_i) = \sum_{n=0}^k t^n (-1)^n E_n(z). \quad (9)$$

Replacing t by $1/t$ and multiplying both sides by t^k , we find

$$\prod_{i=1}^k (t - z_i) = \sum_{n=0}^k t^{k-n} (-1)^n E_n(z) = \sum_{m=0}^k t^m (-1)^{k-m} E_{k-m}(z), \quad (10)$$

of which (6.2-14) is a special case.

EXAMPLE 6.6-3 (Stirling numbers) Similarities between differential calculus and the calculus of finite differences come from the analogy

between the differential operator and the difference operator Δ defined by $\Delta f(x) = f(x+1) - f(x)$. For every positive integer n , there is a pair of corresponding quantities (see Exercise 5.5-1)

$$\left(\frac{d}{dx}\right)^n f(x) = f^{(n)}(x) \quad \text{and} \quad \Delta^n f(x) = F^{(n)}(1, \dots, 1; x, x+1, \dots, x+n). \quad (11)$$

A second pair is

$$x^n \quad \text{and} \quad [x]_n = x(x-1) \cdots (x-n+1) = (-1)^n(-x, n), \quad (12)$$

which satisfy

$$\frac{dx^n}{dx} = nx^{n-1} \quad \text{and} \quad \Delta[x]_n = n[x]_{n-1}. \quad (13)$$

The Stirling numbers of the first and second kinds, $S_n^{(m)}$ and $\mathfrak{S}_n^{(m)}$, are defined as the coefficients in the relations connecting the monomials and their analogs:

$$[x]_n = \sum_{m=0}^n S_n^{(m)} x^m \quad \text{and} \quad x^n = \sum_{m=0}^n \mathfrak{S}_n^{(m)} [x]_m. \quad (14)$$

We use the definition of Stirling numbers adopted by Abramowitz (1964).

Defining $[x]_0 = 1$, we observe that $S_0^{(0)} = \mathfrak{S}_0^{(0)} = 1$, and henceforth we assume $n \geq 1$. Comparison of (10) and the first equation of (14) shows that

$$\begin{aligned} S_n^{(m)} &= (-1)^{n-m} E_{n-m}(0, 1, \dots, n-1) \\ &= (-1)^{n-m} \binom{n}{m} R_{n-m}(-1, \dots, -1; 0, 1, \dots, n-1). \end{aligned} \quad (15)$$

The second equation of (14) is a special case of Newton's divided-difference interpolation formula. In (5.5-7) we set $f(x) = x^n$, $f^{(m)}(x)/m! = \binom{n}{m} x^{n-m}$, $p = n+1$, and $z_i = i-1$, $i = 1, 2, \dots, n$. Then

$$x^n = \sum_{m=1}^n \binom{n}{m} R_{n-m}(1, \dots, 1; 0, 1, \dots, m) x(x-1) \cdots (x-m+1). \quad (16)$$

Upon comparing with (14) and using (6.2-11), we have

$$\mathfrak{S}_n^{(m)} = \binom{n}{m} R_{n-m}(1, \dots, 1; 0, 1, \dots, m) = C_{n-m}(0, 1, \dots, m). \quad (17)$$

The last member is equal to $C_{n-m}(1, \dots, m)$ provided $m \geq 1$. Similarly, the zero argument of E_{n-m} in (15) can be omitted if $n > 1$ and $m \geq 1$.

A generating relation for $\mathfrak{S}_n^{(m)}$ is apparent from (17) (see Exercise 6.6-2).

Let $n \in \mathbb{N}$ and suppose that f is holomorphic on an open disk with center x and radius greater than n . By (11), (17), and (5.7-4), the differences and derivatives of f are related by

$$\frac{1}{n!} \Delta^n f(x) = \sum_{m=n}^{\infty} \mathfrak{S}_m^{(n)} \frac{1}{m!} f^{(m)}(x). \quad (18)$$

EXAMPLE 6.6-4 (Fibonacci numbers and Lucas numbers) Let the roots of the equation $t^2 = t + 1$ be denoted by

$$x = \frac{1 + \sqrt{5}}{2} = 1.618 \dots, \quad y = \frac{1 - \sqrt{5}}{2} = -0.618 \dots \quad (19)$$

Note that $x + y = 1$ and $xy = -1$. For every integer n define the Fibonacci number F_n and the Lucas number L_n by

$$F_n = \frac{x^n - y^n}{x - y}, \quad L_n = x^n + y^n. \quad (20)$$

The equation $x^2 = x + 1$ implies $x^{n+1} = x^n + x^{n-1}$, and similarly for y . Hence,

$$F_{n+1} = F_n + F_{n-1}, \quad L_{n+1} = L_n + L_{n-1}. \quad (21)$$

The Fibonacci and Lucas numbers are two linearly independent solutions with different initial conditions of the same second-order difference equation. Since $y = -1/x$ implies $F_{-n} = (-1)^{n+1} F_n$ and $L_{-n} = (-1)^n L_n$, we need to consider only the nonnegative integers n . By (20) and (21),

$$\begin{aligned} F_0, F_1, F_2, \dots &= 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, \dots, \\ L_0, L_1, L_2, \dots &= 2, 1, 3, 4, 7, 11, 18, 29, 47, 76, 123, 199, 322, \dots \end{aligned} \quad (22)$$

From (5.5-3) and (6.2-17) we find

$$\begin{aligned} F_n &= nR_{n-1} \left(1, 1; \frac{1 + \sqrt{5}}{2}, \frac{1 - \sqrt{5}}{2} \right), \\ L_n &= 2 \lim_{\beta \rightarrow 0} R_n \left(\beta, \beta; \frac{1 + \sqrt{5}}{2}, \frac{1 - \sqrt{5}}{2} \right). \end{aligned} \quad (23)$$

Note that F_n is a complete symmetric function (Section 6.2) of degree $n - 1$ in x and y . Either (1) or (8) implies the generating relation

$$(1 - t - t^2)^{-1} = \sum_{n=0}^{\infty} t^n F_{n+1}, \quad |t| < |y|, \quad (24)$$

which can be obtained alternatively by subtracting the Taylor series of

$(1 - ty)^{-1}$ from that of $(1 - tx)^{-1}$. Addition of these Taylor series yields

$$\frac{2 - t}{1 - t - t^2} = \sum_{n=0}^{\infty} t^n L_n, \quad |t| < |y|. \quad (25)$$

Another generating relation for L_n ,

$$-\log(1 - t - t^2) = \sum_{n=1}^{\infty} \frac{t^n}{n} L_n, \quad |t| < |y|, \quad (26)$$

is the sum of the Taylor series of $-\log(1 - tx)$ and $-\log(1 - ty)$.

If we define

$$r_n = F_{n+1}/F_n, \quad n - 1 \in \mathbb{N}, \quad (27)$$

then (21) implies

$$r_n = 1 + 1/r_{n-1}, \quad n - 2 \in \mathbb{N}. \quad (28)$$

Since $x > |y|$, it follows from (20) that

$$\lim_{n \rightarrow \infty} r_n = x = \frac{1 + \sqrt{5}}{2}. \quad (29)$$

In a book published in 1202 Leonardo da Pisa (nicknamed Fibonacci) included an arithmetical exercise in which the number of offspring of a pair of rabbits grows according to the first sequence in (22). The F_n have many entertaining associations, for example with some simple networks of electrical resistances (Exercise 6.6-5) and with the arrangement of florets in a sunflower and of scales in a fir cone [see Coxeter (1953) and Hoggatt (1969)]. Iteration of (28) leads to developments in continued fractions [Vorob'ev (1961)]. Since $x/1 = (x + 1)/x$, (29) connects the r_n with Euclid's golden section of a line segment into two pieces with lengths in the ratio $x/1$ such that the whole segment $(x + 1)$ has the same proportion to the longer piece (x) as does the longer to the shorter piece. A rectangle with sides in this ratio is said to be golden, and the front of the Parthenon apparently is a golden rectangle surmounted by a triangle. The same ratio occurs in the star pentagram (the figure formed by the diagonals of a regular pentagon), which had mystical significance for the Pythagoreans.

In operations research Fibonacci numbers occur in a method (often called Fibonacci search) for finding the maximum of a function of one variable [Kiefer (1953)]. A simplified version known as search by golden section or the 0.618 method is often applied to industrial problems in mainland China.

6.7 Polynomials of Legendre, Gegenbauer, and Chebyshev

The Gegenbauer polynomials, often called ultraspherical polynomials, include Legendre and Chebyshev polynomials as special or limiting cases. Gegenbauer polynomials are closely connected with axially symmetric potentials in n dimensions, and we begin with the case $n = 3$, which corresponds to Legendre polynomials.

In three-dimensional Euclidean space the gravitational potential at the point $\mathbf{r}$ due to a massive particle at $\mathbf{r}' \neq \mathbf{r}$ is a constant multiple of

$$\frac{1}{|\mathbf{r} - \mathbf{r}'|} = (r^2 - 2rr' \cos \theta + r'^2)^{-1/2}, \quad (1)$$

where r and r' are the radial coordinates of the two points and θ is their angular separation. The same expression gives the electric potential at $\mathbf{r}$ due to a charged particle at $\mathbf{r}'$. In celestial mechanics, electrostatics, and atomic physics, it is often useful to expand this potential in powers of r/r' or r'/r , whichever is smaller. Define $r_< = \min\{r, r'\}$, $r_> = \max\{r, r'\}$, and $t = r_>/r_<$, so that

$$\frac{1}{|\mathbf{r} - \mathbf{r}'|} = \frac{1}{r_>} (1 - 2t \cos \theta + t^2)^{-1/2}. \quad (2)$$

The Legendre polynomials are defined to be the coefficients of the Taylor series

$$(1 - 2t \cos \theta + t^2)^{-1/2} = \sum_{n=0}^{\infty} t^n P_n(\cos \theta), \quad (3)$$

where $\cos \theta$ may be real or complex. It is easy to verify that

$$P_0 = 1, \quad P_1(x) = x, \quad P_2(x) = \frac{3}{2}x^2 - \frac{1}{2}, \quad P_3(x) = \frac{5}{2}x^3 - \frac{3}{2}x. \quad (4)$$

Since $1 - 2t \cos \theta + t^2 = (1 - te^{i\theta})(1 - te^{-i\theta})$, comparison with (6.6-1) shows that

$$P_n(\cos \theta) = R_n\left(\frac{1}{2}, \frac{1}{2}; e^{i\theta}, e^{-i\theta}\right),$$

$$P_n(x) = R_n\left[\frac{1}{2}, \frac{1}{2}; x + (x^2 - 1)^{1/2}, x - (x^2 - 1)^{1/2}\right]. \quad (5)$$

Equation (5.9-25) gives the recurrence relation

$$(n+1)P_{n+1}(x) - (2n+1)xP_n(x) + nP_{n-1}(x) = 0, \quad (6)$$

which makes it easy to continue the listing (4).

The Taylor series (3) has a circle of convergence passing through the branch point closest to 0. Since the branch points are given by

$$t = e^{\pm i\theta}, \quad \cos \theta = \frac{1}{2}(t + t^{-1}), \quad (7)$$

6.7 Polynomials of Legendre, Gegenbauer, and Chebyshev 149

the series converges if

$$|t| < e^{-|\operatorname{Im} \theta|} \quad (8)$$

and diverges if the inequality is reversed. Note that the radius of the disk of convergence in the t plane can never exceed 1.

If t is fixed with $|t| < 1$, (8) defines an open region of convergence in the $(\cos \theta)$ plane with boundary passing through the branch point given by the second equation of (7). The equation of the boundary is

$$e^{-|\operatorname{Im} \theta|} = |t|. \quad (9)$$

To see what curve in the $(\cos \theta)$ plane corresponds to a constant value of $\operatorname{Im} \theta$, set

$$\begin{aligned} \cos \theta &= \cos(\theta' + i\theta'') = \cos \theta' \cosh \theta'' + i \sin \theta' \sinh \theta'' = u + iv, \\ \left(\frac{u}{\cosh \theta''}\right)^2 + \left(\frac{v}{\sinh \theta''}\right)^2 &= \cos^2 \theta' + \sin^2 \theta' = 1. \end{aligned} \quad (10)$$

Hence the curve is an ellipse with semiaxes of lengths

$$\cosh(\operatorname{Im} \theta) = \frac{1}{2}(|t|^{-1} + |t|), \quad \sinh|\operatorname{Im} \theta| = \frac{1}{2}(|t|^{-1} - |t|) \quad (11)$$

and foci $-1, 1$. The ellipse passes through the branch point $\frac{1}{2}(t + t^{-1})$ of the generating function, and the series (3) converges inside the ellipse and diverges outside. Convergence at points on the ellipse is a more delicate matter, like convergence of a power series at points on the circle of convergence. As $|t| \rightarrow 1$ the ellipse collapses onto the interfocal line segment and no longer has an inner region where convergence is guaranteed.

According to (5.4-15) the axially symmetric potential which reduces to r^n , $n \in \mathbb{N}$, when $\theta = 0$ is

$$R_n\left(\frac{1}{2}, \frac{1}{2}; r \cos \theta + ir \sin \theta, r \cos \theta - ir \sin \theta\right) = r^n P_n(\cos \theta). \quad (12)$$

Likewise, the potential which reduces to r^{-n-1} when $\theta = 0$ is

$$r^{-n-1} R_{-n-1}\left(\frac{1}{2}, \frac{1}{2}; e^{i\theta}, e^{-i\theta}\right), \quad r \neq 0. \quad (13)$$

(Strictly speaking, we should require $\cos \theta > 0$ until the definition of the R function is extended in the next section.) By (5.9-21), (13) is equal to

$$r^{-n-1} P_n(\cos \theta). \quad (14)$$

Hence the general term of the series expansion [see (2)]

$$\frac{1}{|\mathbf{r} - \mathbf{r}'|} = \sum_{n=0}^{\infty} \frac{r_{>}^n}{r_{>}^{n+1}} P_n(\cos \theta), \quad r \neq r', \quad (15)$$

is a potential of the type (12) or (14) according as $r < r'$ or $r > r'$.

150 6.7 Polynomials of Legendre, Gegenbauer, and Chebyshev

In $(p+2)$ -dimensional Euclidean space, $p = 1, 2, 3, \dots$, axially symmetric potentials corresponding to (12) and (14) are found by setting $x_n = r \cos \theta$ and $\rho = r \sin \theta$ in (5.4-13) and again using (5.9-21). They are

$$r^n R_n(p/2, p/2; e^{i\theta}, e^{-i\theta}) \quad (16)$$

and

$$r^{-n-1} R_{-n-1}(p/2, p/2; e^{i\theta}, e^{-i\theta}) = r^{-n-1} R_{n+1-p}(p/2, p/2; e^{i\theta}, e^{-i\theta}). \quad (17)$$

A particular case of (17) is r^{-p} . Since this satisfies the $(p+2)$ -dimensional Laplace equation if $r \neq 0$, so by shift of the origin does $|\mathbf{r} - \mathbf{r}'|^{-p}$ if $\mathbf{r} \neq \mathbf{r}'$. Expanding in R polynomials by (6.6-1),

$$\begin{aligned} |\mathbf{r} - \mathbf{r}'|^{-p} &= r_{>}^{-p} (1 - 2t \cos \theta + t^2)^{-p/2} \\ &= \sum_{n=0}^{\infty} \frac{r_{<}^n}{r_{>}^{n+p}} \frac{(p, n)}{n!} R_n(p/2, p/2; e^{i\theta}, e^{-i\theta}). \end{aligned} \quad (18)$$

For every positive integer p the series converges if $(r_{<}/r_{>}) < \exp(-|\operatorname{Im} \theta|)$ and diverges if the inequality is reversed. The n th term is a potential of the type (16) or (17) according as $r < r'$ or $r > r'$.

If ν is any complex number, the Gegenbauer polynomials of order ν are defined by the generating relation

$$(1 - 2t \cos \theta + t^2)^{-\nu} = \sum_{n=0}^{\infty} t^n C_n^{\nu}(\cos \theta), \quad |t| < e^{-|\operatorname{Im} \theta|}. \quad (19)$$

It is easy to verify that

$$C_0^{\nu} = 1, \quad C_1^{\nu}(x) = 2\nu x, \quad C_2^{\nu}(x) = 2\nu(\nu+1)x^2 - \nu. \quad (20)$$

Comparison of (19) and (6.6-1) shows that, for every $n \in \mathbb{N}$,

$$\begin{aligned} C_n^{\nu}(\cos \theta) &= \frac{(2\nu, n)}{n!} R_n(\nu, \nu; e^{i\theta}, e^{-i\theta}), \\ C_n^{\nu}(x) &= \frac{(2\nu, n)}{n!} R_n[\nu, \nu; x + (x^2 - 1)^{1/2}, x - (x^2 - 1)^{1/2}]. \end{aligned} \quad (21)$$

By (5.9-25), for every positive integer n ,

$$(n+1)C_{n+1}^{\nu}(x) - 2(\nu+n)x C_n^{\nu}(x) + (2\nu+n-1)C_{n-1}^{\nu}(x) = 0. \quad (22)$$

It follows that $C_n^{\nu}(x)$ is a polynomial in ν and x , of degree n in each. Equations (19)–(22) generalize Equations (3)–(6) for the Legendre polynomials $P_n = C_n^{1/2}$. By (16) and (17), with n replaced by $n+p-1$ in (17),

6.7 Polynomials of Legendre, Gegenbauer, and Chebyshev 151

$r^n C_n^{p/2}(\cos \theta)$ and $r^{-n-p} C_n^{p/2}(\cos \theta)$ are axially symmetric potentials in $p + 2$ dimensions for $p - 1 \in \mathbb{N}$ and $n \in \mathbb{N}$. Equation (18) can be written as

$$|\mathbf{r} - \mathbf{r}'|^{-p} = \sum_{n=0}^{\infty} \frac{r_{<}^n}{r_{>}^{n+p}} C_n^{p/2}(\cos \theta), \quad p - 1 \in \mathbb{N}. \quad (23)$$

If $-v \in \mathbb{N}$, $C_n^v = 0$ for $n > -2v$ by (19). If $v \in U$ Theorem 6.6-2 shows that

$$\limsup_{n \rightarrow \infty} |C_n^v(\cos \theta)|^{1/n} = e^{|\operatorname{Im} \theta|}. \quad (24)$$

Hence the convergence (at least insofar as it is determined by the root test) of a series of Gegenbauer polynomials with $v \in U$ is the same as for Legendre polynomials.

Any R polynomial with $k = 2$ and $b_1 = b_2$ can be written as a Gegenbauer polynomial. To do so, we observe that the arguments on the right side of (21) are reciprocals of each other even if θ is complex, and $\cos \theta$ is half their sum. For any complex x and y ,

$$R_n(v, v; x, y) = (xy)^{n/2} R_n[v, v; (x/y)^{1/2}, (y/x)^{1/2}], \quad (25)$$

and hence

$$\frac{(2v, n)}{n!} R_n(v, v; x, y) = (xy)^{n/2} C_n^v \left(\frac{x + y}{2(xy)^{1/2}} \right). \quad (26)$$

From Theorem 6.2-5 it follows that

$$\lim_{v \rightarrow 0} R_n(v, v; x, y) = \frac{1}{2} (x^n + y^n) \quad (27)$$

and hence

$$\lim_{v \rightarrow 0} R_n(v, v; e^{i\theta}, e^{-i\theta}) = \cos n\theta. \quad (28)$$

The Chebyshev polynomials are defined for $n \in \mathbb{N}$ by

$$\begin{aligned} T_n(\cos \theta) &= \cos n\theta = \frac{1}{2}(e^{in\theta} + e^{-in\theta}) \\ T_n(x) &= \cos(n \arccos x) = \frac{1}{2}[x + (x^2 - 1)^{1/2}]^n + \frac{1}{2}[x - (x^2 - 1)^{1/2}]^n. \end{aligned} \quad (29)$$

Consequently,

$$T_n(\cos \theta) = \lim_{v \rightarrow 0} R_n(v, v; e^{i\theta}, e^{-i\theta}) = \lim_{v \rightarrow 0} \frac{C_n^v(\cos \theta)}{C_n^v(1)}. \quad (30)$$

By (5.9-25) or the addition theorem for the cosine function, we find the recurrence relation

$$T_{n+1}(x) - 2xT_n(x) + T_{n-1}(x) = 0, \quad (31)$$

152 6.7 Polynomials of Legendre, Gegenbauer, and Chebyshev

whereupon it is easy to extend the listing

$$T_0 = 1, \quad T_1(x) = x, \quad T_2(x) = 2x^2 - 1, \quad T_3(x) = 4x^3 - 3x. \quad (32)$$

Note that the coefficient of x^n in $T_n(x)$ is $2^{n-1}(1 + \delta_{n0})$.

Two generating relations for the Chebyshev polynomials follow easily from the geometric series

$$(1 - te^{i\theta})^{-1} = \sum_{n=0}^{\infty} t^n e^{in\theta}, \quad |t| < e^{\operatorname{Im} \theta}.$$

Changing the sign of θ and adding the two series, we find

$$\frac{1 - t \cos \theta}{1 - 2t \cos \theta + t^2} = \sum_{n=0}^{\infty} t^n T_n(\cos \theta), \quad |t| < e^{-|\operatorname{Im} \theta|}. \quad (33)$$

Multiplying (33) by two and subtracting unity, we have

$$\frac{1 - t^2}{1 - 2t \cos \theta + t^2} = 1 + 2 \sum_{n=1}^{\infty} t^n T_n(\cos \theta), \quad |t| < e^{-|\operatorname{Im} \theta|}. \quad (34)$$

This generating function is particularly interesting because it is the Poisson kernel

$$P_t(\theta) = \frac{1 - t^2}{1 - 2t \cos \theta + t^2} = \sum_{n=-\infty}^{\infty} t^{|n|} e^{in\theta}, \quad |t| < e^{-|\operatorname{Im} \theta|}. \quad (35)$$

When θ is real, this kernel has the important property [see Rudin (1974, Chapter 11)] that

$$f(re^{i\theta}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(e^{i\varphi}) P_r(\theta - \varphi) d\varphi, \quad 0 \leq r < 1, \quad (36)$$

for any function f which is harmonic on the open unit disk and continuous on the closed disk. Conversely, given a function f which is continuous on the unit circle, the right side of (36) is the solution of Dirichlet's problem: to find a harmonic function on the unit disk whose boundary values are f .

Since the left side of either (33) or (34) has poles in the t plane at $e^{i\theta}$ and $e^{-i\theta}$, the radius of convergence of its Taylor series is $\rho = e^{-|\operatorname{Im} \theta|}$. Applying the root test to the series on the right side, we find $\rho^{-1} = \limsup |T_n(\cos \theta)|^{1/n}$. Hence

$$\limsup_{n \rightarrow \infty} |T_n(\cos \theta)|^{1/n} = e^{|\operatorname{Im} \theta|}. \quad (37)$$

By (29) the upper limit is a limit if $\operatorname{Im} \theta \neq 0$. Comparison of (37) and (24) shows that a series of Chebyshev polynomials has the same region of convergence as a series of Legendre or Gegenbauer polynomials.

The Chebyshev polynomials have acquired great practical importance in numerical analysis since the advent of automatic computation. We shall say a few words about this after the polynomials have been given a different normalization in Section 6.10.

6.8 Analytic continuation of the R function

We proved in Corollary 6.3-4 that $R_t(b, z)/\Gamma(c)$ is an entire function of the parameters t and b if the variables $z_1, \dots, z_k$ are restricted to an open half-plane which does not contain the origin. Our aim in this section is to enlarge the half-plane to a plane cut along the nonpositive real axis, which we denote by

$$\mathbb{C}_0 = \{x \in \mathbb{C} : x \neq 0, |\operatorname{ph} x| < \pi\}. \quad (1)$$

Since x^t is holomorphic on $\mathbb{C}_0$, it is natural to encounter $\mathbb{C}_0$ in the continuation of R_t . If the points $z_1, \dots, z_k$ lie in $\mathbb{C}_0$, their convex hull may contain the origin, and in this case the representations of $R_t(b, z)$ as a multiple integral (5.9-1) or an infinite series (5.7-4) do not exist. They are supplanted by a remarkable representation as a single integral (subject to restrictions on t and b). This representation accounts for much of the practical importance of the R function, for example its applications to elliptic integrals. Even the restrictions on t and b can be removed by integrating around a contour rather than along an interval of the real line, and the contour integral shows the full extent to which the R function can be holomorphically continued.

THEOREM 6.8-1 Let $\alpha = (a, a') \in \mathbb{C}_+^2$ and $b \in \mathbb{C}^k$, and assume $a + a' = \sum_{i=1}^k b_i$. Let μ_α be a Dirichlet measure on $[0, 1]$. If $z \in \mathbb{C}_0^k$ choose $|\operatorname{ph}(1 - u + uz_i)| < \pi$ for $0 \leq u \leq 1$ and $i = 1, \dots, k$. Then $R_{-a}(b, z)$ has a holomorphic continuation in z to $\mathbb{C}_0^k$, where it is represented by

$$R_{-a}(b, z) = \int_0^1 \prod_{i=1}^k (1 - u + uz_i)^{-b_i} d\mu_\alpha(u). \quad (2)$$

Remark If $c = \sum_{i=1}^k b_i$, then $c = a + a'$ and $\operatorname{Re} c > \operatorname{Re} a > 0$.

Proof Equation (6.3-6) represents the left side of (2) if $|1 - z| < 1$, and we will first prove (2) with this restriction. By (6.6-1),

$$\prod_{i=1}^k (1 - u + uz_i)^{-b_i} = \sum_{n=0}^{\infty} u^n \frac{(c, n)}{n!} R_n(b, 1 - z). \quad (3)$$

The series converges if $|u| < |1 - z|^{-1}$ and therefore converges uniformly for $0 \leq u \leq 1$. Integrating term by term and using (4.4-8), we find

$$\int_0^1 \prod_{i=1}^k (1 - u + uz_i)^{-b_i} d\mu_a(u) = \sum_{n=0}^{\infty} \frac{(a, n)}{n!} R_n(b, 1 - z). \quad (4)$$

By (6.3-6) this proves (2) if $|1 - z| < 1$.

We show next that the integral is holomorphic in z on $\mathbb{C}_0^k$ and thus provides explicitly the continuation of $R_{-a}(b, z)$. Referring to Appendix B.4, we choose $T = [0, 1]$, $Z = \mathbb{C}_0^k$, and

$$f(z, u) = \prod_{i=1}^k (1 - u + uz_i)^{-b_i}.$$

Then:

- (i) $[z \mapsto f(z, u)]$ is holomorphic on $\mathbb{C}_0^k$ for every fixed $u \in [0, 1]$ since $1 - u + uz_i$ lies on the line segment joining 1 and z_i ,
- (ii) $[u \mapsto f(z, u)] \in L_1(|\mu_a|)$ for every fixed $z \in \mathbb{C}_0^k$ since the function is continuous on $[0, 1]$,
- (iii) If K is a compact set in $\mathbb{C}_0^k$, f is bounded on $K \times [0, 1]$.

It follows from Theorem B.4 that the integral is holomorphic in z on $\mathbb{C}_0^k$. $\square$

Equation (2) has great practical importance because it allows any integral of the form

$$\int (A + Bx)^\alpha \cdots (M + Nx)^\mu dx, \quad (5)$$

where $A, B, \alpha, \dots, M, N, \mu$ are constants, to be evaluated in terms of the R function. One useful example, which is obtained by setting $u = 1/(t + 1)$ in (2), is

$$R_{-a}(b, z) = \frac{1}{B(a, a')} \int_0^\infty t^{a'-1} \prod_{i=1}^k (t + z_i)^{-b_i} dt, \quad a + a' = \sum_{i=1}^k b_i, \quad (6)$$

where $a, a' \in \mathbb{C}_>$ and $|\text{ph}(t + z_i)| < \pi$ for $t \geq 0$ and $i = 1, \dots, k$. Other explicit formulas are given in Exercise 6.8-8 and Section 8.1.

We have not yet shown that $R_t(b, z)/\Gamma(c)$ is entire in t and b for every z in $\mathbb{C}_0^k$. For this purpose (2) is unsuitable because of the restriction $\alpha \in \mathbb{C}_>^2$. We shall now exhibit a representation by a contour integral which allows maximum possible freedom in choosing t, b, z .

THEOREM 6.8-2 If $c = \sum_{i=1}^k b_i$, the function $R_{-a}(b, z)/\Gamma(c)$ has a holomorphic continuation in (a, b, z) to $\mathbb{C}^{k+1} \times \mathbb{C}_0^k$, where it is represented as follows: Given $z \in \mathbb{C}_0^k$, let γ be a positively oriented ellipse in $\mathbb{C}$ with foci 0 and 1 such that $(1 - z_i)^{-1}$ is outside γ for $i = 1, \dots, k$. [$(1 - z_i)^{-1}$ is considered to be outside γ if $z_i = 1$.] Choose $|\text{ph}(1 - s + sz_i)| < \pi$ for every s on γ and $i = 1, \dots, k$. Then

$$\frac{R_{-a}(b, z)}{\Gamma(c)} = \frac{1}{2\pi i} \int_{\gamma} \frac{R_{-1}(a, c - a; s - 1, s)}{\Gamma(c)} \prod_{i=1}^k (1 - s + sz_i)^{-b_i} ds. \quad (7)$$

Remark Choosing γ to be an ellipse is convenient but inessential by Cauchy's integral theorem. To see that the specified choice of phase is possible, note that s must have a smaller absolute value than $(1 - z_i)^{-1}$ if their arguments are equal. Hence $s/(1 - z_i)^{-1}$ must be less than unity if it is real and positive. Then $1 - s(1 - z_i)$ cannot be real and negative or zero.

Proof The left side of (7) is represented by (2) if $\text{Re } c > \text{Re } a > 0$, and we first prove (7) with this restriction. Define a holomorphic function g on γ and its inner region by

$$g(s) = \prod_{i=1}^k (1 - s + sz_i)^{-b_i}, \quad |\text{ph}(1 - s + sz_i)| < \pi. \quad (8)$$

Note that the branch point $(1 - z_i)^{-1}$ is outside γ . By (2) and (5.2-1),

$$R_{-a}(b, z) = \int_0^1 g(u) d\mu_a(u) = G(a, c - a; 1, 0), \quad (9)$$

where G is the Dirichlet average of g . Since the convex hull of $\{1, 0\}$ is the interfocal line segment of the ellipse γ , we find from (6.3-4) that

$$G(a, c - a; 1, 0) = \frac{1}{2\pi i} \int_{\gamma} g(s) R_{-1}(a, c - a; s - 1, s) ds. \quad (10)$$

By (9) and (10), (7) is proved if $\text{Re } c > \text{Re } a > 0$.

We now fix γ and let z vary. Given $\delta > 0$ let γ be the ellipse

$$|s| + |s - 1| = 1 + \delta \quad (11)$$

described in the positive direction. The integral in (7) is well defined if each z_i belongs to the set of points ζ such that $(1 - \zeta)^{-1}$ is outside γ , i.e.,

$$|(1 - \zeta)^{-1}| + |(1 - \zeta)^{-1} - 1| > 1 + \delta.$$

We multiply by $|1 - \zeta|$ and find that the set in question is

$$\mathbb{C}_{\delta} = \{\zeta \in \mathbb{C}_0 : 1 + |\zeta| > (1 + \delta)|1 - \zeta|\}. \quad (12)$$

Note that $\mathbb{C}_\delta$ tends to $\mathbb{C}_0$ as $\delta \rightarrow 0$. We shall show that the integral in (7) is holomorphic in (a, b, z) on $\mathbb{C}^{k+1} \times \mathbb{C}_\delta^k$ and so provides the explicit holomorphic continuation of the left side. Since δ can be arbitrarily small, the proof of the theorem will then be complete.

Let γ^* be the set of points lying on the oriented contour γ . Referring to Theorem B.4, we choose $T = \gamma^*$, $Z = \mathbb{C}^{k+1} \times \mathbb{C}_\delta^k$, and $d\mu = ds$, which implies $d|\mu| = |ds|$. By Corollary 6.3-5 the integrand in (7) is holomorphic on Z for every fixed $s \in \gamma^*$, continuous on $Z \times \gamma^*$, and bounded on $K \times \gamma^*$ if K is any compact subset of Z . It follows at once that the three assumptions of Theorem B.4 are satisfied and the integral is holomorphic on Z . $\square$

From (7) we can recover (2) by a method which is worth describing briefly [for details see Carlson (1974a)]. Cut the plane along the line segment with endpoints 0 and 1, and deform the ellipse γ (using Cauchy's integral theorem) into a contour consisting of the upper and lower edges of the cut and two small circles of radius ρ centered on 0 and 1. If $\operatorname{Re} c > \operatorname{Re} a > 0$, the integrals around the two circles can be shown to vanish with ρ . Moreover, if

$$f(a, c; s) = R_{-1}(a, c - a; s - 1, s)/\Gamma(c) \quad (13)$$

and if $0 < u < 1$ and $\operatorname{Re} c > \operatorname{Re} a > 0$, then the discontinuity of f across the cut is found to be [see (7.8-5)]

$$\lim_{\varepsilon \rightarrow 0+} [f(a, c; u - i\varepsilon) - f(a, c; u + i\varepsilon)] = 2\pi i \frac{u^{a-1}(1-u)^{c-a-1}}{\Gamma(a)\Gamma(c-a)}. \quad (14)$$

Therefore the two integrals along the edges of the cut can be combined to give (2).

Because the ${}_2F_1$ function can be written as an R function by (5.9-12), it follows from Theorem 6.8-2 that ${}_2F_1(\alpha, \beta; \gamma; x)/\Gamma(\gamma)$ has a holomorphic continuation in $(\alpha, \beta, \gamma, x)$ to $\mathbb{C}^3 \times P$, where $P = \{x \in \mathbb{C} : 1 - x \in \mathbb{C}_0\}$ is the complex plane cut along the positive real line from 1 to ∞ . The continuation found in Exercise 4.4-2 required $\operatorname{Re} \gamma > \operatorname{Re} \beta > 0$.

Since we have succeeded in continuing the R function to $\mathbb{C}_0^k$, one may wonder whether it would be possible in general to continue F to Ω^k if f is holomorphic on Ω but Ω is not convex as assumed in Theorem 6.3-6. This can indeed be done if Ω is simply connected, but the proof [Carlson (1969)] is more difficult than it is for the R function. Since F was originally defined as an average of f over $\operatorname{con}(z)$, it is noteworthy that F can have a meaning when $\operatorname{con}(z)$ may lie partly outside a natural boundary of f .

The R function in two variables can be transformed by Proposition 5.9-8 so that any one of four parameters becomes the degree of homogeneity, but in more than two variables there are only two choices instead of four. We prove now a very useful generalization of (5.9-21). The set U_k was defined in (6.3-2).

THEOREM 6.8-3 (Euler's transformation) Let $a \in \mathbb{C}$ and $b \in U_k$, and define a' by $a + a' = \sum_{i=1}^k b_i$. Let $z \in \mathbb{C}_0^k$ and define $z^{-1} = (z_1^{-1}, \dots, z_k^{-1})$. Then

$$R_{-a}(b, z) = \prod_{i=1}^k z_i^{-b_i} R_{-a'}(b, z^{-1}). \quad (15)$$

Remark . If $k = 2$, we recover (5.9-21) by using homogeneity and symmetry. If $a' = 0$ we recover (6.6-5) with z less restricted.

Proof Suppose first that $a, a' \in \mathbb{C}_>$ and set $u = 1 - v$ in (2). Then $d\mu_\alpha(u) = -d\mu_{\alpha'}(v)$, where $\alpha = (a, a')$ and $\alpha' = (a', a)$. Moreover,

$$1 - u + uz_i = z_i[(1 - u)z_i^{-1} + u] = z_i(1 - v + vz_i^{-1}),$$

where $\text{ph}(1 - v + vz_i^{-1})$ changes continuously from 0 at $v = 0$ to $-\text{ph } z_i$ at $v = 1$. Thus (2) implies (15) if $a, a' \in \mathbb{C}_>$. This condition can now be dropped by the permanence of functional relations, since both sides of (15) are holomorphic in (a, b) on $\mathbb{C} \times U_k$ by Theorem 6.8-2. $\square$

Now that the R function has been holomorphically continued, we can define the Legendre function of any complex degree α by

$$P_\alpha(\cos \theta) = R_\alpha\left(\frac{1}{2}, \frac{1}{2}; e^{i\theta}, e^{-i\theta}\right), \quad |\text{Re } \theta| < \pi. \quad (16)$$

This agrees with (6.7-5) if $\alpha \in \mathbb{N}$. More generally the Gegenbauer function of any complex degree α and any complex order ν such that $2\nu + \alpha \in U$ is defined by

$$C_\alpha^\nu(\cos \theta) = \frac{\Gamma(2\nu + \alpha)}{\Gamma(1 + \alpha)\Gamma(2\nu)} R_\alpha(\nu, \nu; e^{i\theta}, e^{-i\theta}), \quad |\text{Re } \theta| < \pi. \quad (17)$$

This reduces to (16) if $\nu = \frac{1}{2}$ and to (6.7-21) if $\alpha \in \mathbb{N}$. Any R function with $k = 2$ and $b_1 = b_2$ can be written as a Gegenbauer function by an extension of (6.7-26),

$$\frac{1}{\Gamma(2\nu)} R_\alpha(\nu, \nu; x, y) = \frac{\Gamma(1 + \alpha)}{\Gamma(2\nu + \alpha)} (xy)^{\alpha/2} C_\alpha^\nu \left[\frac{x + y}{2(xy)^{1/2}} \right], \quad (18)$$

where $x, y \in \mathbb{C}_0$. Theorem 6.8-2 ensures that the left side is entire in α and ν , but we shall see now that this statement can be strengthened.

THEOREM 6.8-4 Let $b = (\beta, \beta, \dots, \beta) \in \mathbb{C}^k$ and $z \in \mathbb{C}_0^k$. Then $\Gamma(\beta)R_i(b, z)/\Gamma(k\beta)$ is holomorphic in (t, β, z) on $\mathbb{C}^2 \times \mathbb{C}_0^k$.

Remark It follows by (3.7-2) that $R_\alpha(v, v; x, y)/\Gamma(v + \frac{1}{2})$ is holomorphic in (α, v, x, y) on $\mathbb{C}^2 \times \mathbb{C}_0^2$.

Proof For every $(t, z) \in \mathbb{C} \times \mathbb{C}_0^k$ it is apparent from Corollary 6.3-7 [with $\Omega = \mathbb{C}_>$ and $f(x) = x^t$] that $R_i(b, z)/\Gamma(k\beta) = 0$ if $-\beta \in \mathbb{N}$. By the permanence of functional relations the equality holds for every $(t, z) \in \mathbb{C} \times \mathbb{C}_0^k$ if $-\beta \in \mathbb{N}$.

By Theorems 6.8-2 and 3.3-1, $\Gamma(\beta)R_i(b, z)/\Gamma(k\beta)$ is holomorphic in (t, β, z) on $\mathbb{C} \times U \times \mathbb{C}_0^k$. If $n \in \mathbb{N}$, $\Gamma(\beta)$ has a simple pole at $\beta = -n$, while $R_i(b, z)/\Gamma(k\beta)$ is entire in β and has a zero at $\beta = -n$. Hence their product tends to a finite limit as $\beta \rightarrow -n$, and the singularity is removable. $\square$

The Gegenbauer function is the same except for multiplicative factors as the associated Legendre function [Erdélyi (1953), 3.2(29)],

$$P_v^\mu(x) = \frac{2^\mu (x^2 - 1)^{-\mu/2}}{\Gamma(1 - \mu)} \cdot R_{v+\mu} \left[\frac{1}{2} - \mu, \frac{1}{2} - \mu; x + (x^2 - 1)^{1/2}, x - (x^2 - 1)^{1/2} \right], \quad (19)$$

where $x^2 - 1 = (x + 1)(x - 1)$ and $|\text{ph}(x \pm 1)| < \pi$. This function is widely used because it satisfies Legendre's differential equation (5.9-18), but its properties are complicated by an extra branch point at 1 introduced by the factor $(x - 1)^{-\mu/2}$. If x lies in a plane cut along the negative real axis from -1 to $-\infty$, then $x \pm (x^2 - 1)^{1/2}$ is in $\mathbb{C}_0$, as we see by setting $x = \cos(\theta' + i\theta'')$ where θ' is not an odd multiple of π . Since the R function in (19) is symmetric in the two square roots, which are interchanged when x encircles 1, it is holomorphic in x on the cut plane. Because of the extra branch point at 1, P_v^μ is made holomorphic by extending the cut from -1 to 1. If $-1 < x < 1$, P_v^μ is defined separately by (19) with $(x - 1)^{-\mu/2}$ replaced by $(1 - x)^{-\mu/2}$ [see also (6.10-18)].

The Legendre function of the second kind [Erdélyi (1953, 3.2(36))] is

$$Q_v^\mu(x) = e^{i\mu\pi} \frac{2^v \Gamma(1 + v) \Gamma(1 + v + \mu)}{\Gamma(2 + 2v)} (x^2 - 1)^{-\mu/2} \cdot R_{\mu-v-1}(1 + v, 1 + v; x + 1, x - 1). \quad (20)$$

In the important case $v - \mu \in \mathbb{N}$ (for the general case see Exercise 6.3-1), the R function is an averaged Cauchy kernel (Section 5.11 and Corollary 6.3-5) and is holomorphic on the x plane cut along the line segment

joining -1 and 1 . If $\mu \notin \mathbb{Z}$ the factor $(x^2 - 1)^{-\mu/2}$ has a third branch point at infinity, and Q_v^μ is made holomorphic by extending the cut from -1 to $-\infty$.

The main point is that Gegenbauer functions, Legendre functions, and $R_i(\beta, \beta; x, y)$ are essentially the same. In Sections 6.9 and 6.10 we shall show that any function of this type admits two quadratic transformations.

6.9 The first quadratic transformation

In Section 6.5 we found a transformation (applicable to any R polynomial) in which the new arguments are linear functions of the old ones. In the present section we shall deduce a transformation which is called quadratic because the old and new arguments are related by equations of second degree. The transformation can be applied to any R function of two variables with equal b parameters.

An essential ingredient of the proof is the vanishing of $R_{2n+1}(\beta, \beta; x, -x)$. By homogeneity this function is a multiple of x^{2n+1} and therefore it is odd in x . Since the b parameters are equal, it is symmetric in x and $-x$ and so is even in x . Being both odd and even, it must vanish identically.

This result has important consequences also for confluent hypergeometric functions which we shall discuss before proceeding to the quadratic transformation of an R function. We shall find that an S function in two variables with equal b parameters can be written as a ${}_0F_1$ series (i.e. a Bessel function), and therefore any ${}_0F_1$ series is expressible as a restricted ${}_1F_1$ series.

THEOREM 6.9-1 Let $n \in \mathbb{N}$ and $\beta, x \in \mathbb{C}$. Assume $(\beta + \frac{1}{2}, n) \neq 0$. Then

$$R_{2n+1}(\beta, \beta; x, -x) = 0, \quad (1)$$

$$R_{2n}(\beta, \beta; x, -x) = \frac{(\frac{1}{2}, n)}{(\beta + \frac{1}{2}, n)} x^{2n}. \quad (2)$$

Proof For the moment assume $\beta \in U$ as well as $(\beta + \frac{1}{2}, n) \neq 0$. By (6.6-1),

$$(1 - tx)^{-\beta}(1 + tx)^{-\beta} = \sum_{m=0}^{\infty} t^m \frac{(2\beta, m)}{m!} R_m(\beta, \beta; x, -x).$$

Comparing this with the binomial series,

$$(1 - t^2 x^2)^{-\beta} = \sum_{n=0}^{\infty} t^{2n} \frac{(\beta, n)}{n!} x^{2n},$$

which contains no odd powers of t , we first deduce (1). Secondly, equating coefficients of t^{2n} and dividing by $(2\beta, 2n) \neq 0$, we have

$$R_{2n}(\beta, \beta; x, -x) = \frac{(\beta, n)(1, 2n)}{(1, n)(2\beta, 2n)} x^{2n}.$$

The duplication formula (2.2-7) yields (2). The condition $\beta \in U$ can now be dropped since $R_m(\beta, \beta; x, y)$ is analytic at $\beta = 0, -1, -2, \dots$ by Theorem 6.2-6. $\square$

The next theorem has consequences for Bessel functions which will be discussed at the end of this section.

THEOREM 6.9-2 Let $x, y \in \mathbb{C}$ and $\beta + \frac{1}{2} \in U$. Then

$$S(\beta, \beta; x, y) = e^{(x+y)/2} {}_0F_1\left[\beta + \frac{1}{2}; (x-y)^2/16\right]. \quad (3)$$

Proof By the permanence of functional relations we may suppose $\beta \in \mathbb{C}_>$ in view of Exercise 6.9-19 and the Remark following Corollary 6.3-7. By (5.8-3),

$$S(\beta, \beta; x, y) = e^{(x+y)/2} S\left(\beta, \beta; \frac{x-y}{2}, \frac{y-x}{2}\right). \quad (4)$$

Since the arguments on the right side are equal and opposite, it suffices to prove (3) for the case $y = -x$. By (5.8-4) and Theorem 6.9-1,

$$\begin{aligned} S(\beta, \beta; x, -x) &= \sum_{m=0}^{\infty} \frac{1}{m!} R_m(\beta, \beta; x, -x) \\ &= \sum_{n=0}^{\infty} \frac{(\frac{1}{2}, n)}{(2n)!(\beta + \frac{1}{2}, n)} x^{2n} = \sum_{n=0}^{\infty} \frac{(x^2/4)^n}{(\beta + \frac{1}{2}, n)n!}. \end{aligned}$$

In the last step we have used the duplication formula (2.2-7). The series on the right side is a ${}_0F_1$ series:

$$S(\beta, \beta; x, -x) = {}_0F_1(\beta + \frac{1}{2}; x^2/4). \quad \square \quad (5)$$

By (5.8-6) we can rewrite (5) as

$${}_1F_1(\beta; 2\beta; 2x) = e^x {}_0F_1(\beta + \frac{1}{2}; x^2/4). \quad (6)$$

This result, known as Kummer's second formula, shows that any ${}_0F_1$ series can be expressed in terms of a restricted ${}_1F_1$ series, as asserted in Section 2.4.

Essentially the same procedure used in proving Theorem 6.9-2 leads to a quadratic transformation of the R function.

TRANSFORMATION 6.9-3 Let $t \in \mathbb{C}$, $\beta + \frac{1}{2} \in U$, and $x, y \in \mathbb{C}_>$. Then

$$R_{2t}(\beta, \beta; x, y) = R_t \left[\beta + t, \frac{1}{2} - t; \left(\frac{x+y}{2} \right)^2, xy \right]. \quad (7)$$

Remark The arguments on the right side are the squares of the arithmetic and geometric means of x and y . Since $x, y \in \mathbb{C}_>$, the means lie in $\mathbb{C}_>$ and their squares in $\mathbb{C}_0$. Equation (7) is a special case of Exercise 6.9-5.

Proof Let $2\beta \in U$ and $|w| < 1$. By (6.3-6), Theorem 6.9-1, and (6.2-5),

$$\begin{aligned} R_{2t}(\beta, \beta; 1+w, 1-w) &= \sum_{m=0}^{\infty} \frac{(-2t, m)}{m!} R_m(\beta, \beta; -w, w) \\ &= \sum_{n=0}^{\infty} \frac{(-2t, 2n)(\frac{1}{2}, n)}{(2n)!(\beta + \frac{1}{2}, n)} w^{2n} = \sum_{n=0}^{\infty} \frac{(-t, n)(\frac{1}{2} - t, n)}{n!(\beta + \frac{1}{2}, n)} w^{2n} \\ &= \sum_{n=0}^{\infty} \frac{(-t, n)}{n!} R_n \left(\frac{1}{2} - t, \beta + t; w^2, 0 \right) \\ &= R_t \left(\frac{1}{2} - t, \beta + t; 1 - w^2, 1 \right). \end{aligned}$$

Assume $|x - y| < |x + y|$ and set $w = (x - y)/(x + y)$. Using the homogeneity property (5.9-3), we obtain (7). By Theorem 6.8-2 the R function on the right side of (7) is holomorphic in its arguments on $\mathbb{C}_0^2$, and hence both sides of (7) are holomorphic in (x, y) on $\mathbb{C}_>^2$. By the permanence of functional relations the restriction $|x - y| < |x + y|$ can be dropped. By Theorem 6.8-4 the assumption $2\beta \in U$ can be relaxed to $\beta + \frac{1}{2} \in U$. $\square$

If $t = n \in \mathbb{N}$, (7) transforms an R polynomial of even degree:

$$R_{2n}(\beta, \beta; x, y) = R_n \left[\beta + n, \frac{1}{2} - n; \left(\frac{x+y}{2} \right)^2, xy \right]. \quad (8)$$

If $(\beta + \frac{1}{2}, n) \neq 0$, this relation between polynomials is valid for all x and y . For an odd polynomial, with $2t = 2n + 1$, we find

$$R_{2n+1}(\beta, \beta; x, y) = R_{n+1/2} \left[\frac{1}{2} + \beta + n, -n; \left(\frac{x+y}{2} \right)^2, xy \right].$$

Despite the nonintegral homogeneity parameter, the right side is a polynomial because the second b parameter is a negative integer. Interchanging these two parameters by (5.9-20), we obtain

$$R_{2n+1}(\beta, \beta; x, y) = \frac{x+y}{2} R_n \left[1 + \beta + n, -\frac{1}{2} - n; \left(\frac{x+y}{2} \right)^2, xy \right]. \quad (9)$$

Note that (9) and (8) reduce to (1) and (2) if $y = -x$.

In the polynomial case the quadratic transformation can of course be followed by a linear transformation. The respective combinations of (8) and (9) with (6.5-1), for example, are

$$\begin{aligned} (2\beta, 2n)R_{2n}(\beta, \beta; x, y) &= (\beta, 2n) \\ &\cdot R_n\left[\frac{1}{2} - \beta - n, \frac{1}{2} - n; (x+y)^2, (x-y)^2\right], \quad (10) \\ (2\beta, 2n+1)R_{2n+1}(\beta, \beta; x, y) &= (\beta, 2n+1)(x+y) \\ &\cdot R_n\left[\frac{1}{2} - \beta - n, -\frac{1}{2} - n; (x+y)^2, (x-y)^2\right]. \end{aligned} \quad (11)$$

Since both sides are polynomials in β, x, y , these two equations hold without exception.

In Section 6.5 we observed that the linear transformation, combined with symmetry, allows any R polynomial in two variables to be written with six choices of arguments. If the b parameters are equal, the quadratic transformation gives a different starting point which leads to six more choices. Each choice corresponds to a different argument of an equivalent ${}_2F_1$ polynomial. If $x = e^{i\theta}$ and $y = e^{-i\theta}$, so that the left sides of (8) and (9) are Gegenbauer polynomials with argument $\cos \theta$, the right sides can be written by (5.9.11) as ${}_2F_1$ polynomials with argument $1 - \cos^2 \theta = \sin^2 \theta$. From (10) and (11) we obtain ${}_2F_1$ polynomials with arguments $1 - \cos^2 \theta / (-\sin^2 \theta) = \csc^2 \theta$. The other four possibilities (see Exercise 6.5-5) are $\cos^2 \theta$, $\sec^2 \theta$, $-\tan^2 \theta$, and $-\cot^2 \theta$.

EXAMPLE 6.9-4 (*Logarithms, inverse circular functions, and inverse hyperbolic functions*) The quadratic transformation (7) contains the elementary connection between logarithms and inverse circular functions. Setting $t = -\frac{1}{2}$ and $\beta = 1$ in (7) and using Exercise 5.9-13, we find

$$\frac{\log x - \log y}{x - y} = R_{-1}(1, 1; x, y) = R_{-1/2}\left[\frac{1}{2}, 1; \left(\frac{x+y}{2}\right)^2, xy\right]. \quad (12)$$

The last R function is related to the inverse circular functions, as one sees by setting $t = xu^{1/2}$ in

$$\begin{aligned} \arcsin x &= \int_0^x (1-t^2)^{-1/2} dt = \frac{x}{2} \int_0^1 u^{-1/2} (1-ux^2)^{-1/2} du \\ &= xR_{-1/2}\left(\frac{1}{2}, 1; 1-x^2, 1\right). \end{aligned} \quad (13)$$

The importance of this function makes it convenient to define

$$R_C(x, y) = R_{-1/2}\left(\frac{1}{2}, 1; x, y\right), \quad (14)$$

the subscript C referring to a circular arc (cf. Section 9.2), which is rectified by the inverse sine. By (13) and (12),

$$\begin{aligned}\arcsin x &= xR_C(1 - x^2, 1), \\ R_C(x, y) &= (y - x)^{-1/2} \arccos(x/y)^{1/2},\end{aligned}\quad (15)$$

$$\begin{aligned}\log x &= (x - 1)R_C\left[\left(\frac{x + 1}{2}\right)^2, x\right], \\ R_C(x, y) &= (x - y)^{-1/2} \log \frac{x^{1/2} + (x - y)^{1/2}}{y^{1/2}}.\end{aligned}\quad (16)$$

By simple substitutions in (15) one can express any inverse circular or inverse hyperbolic function in terms of R_C (Exercise 6.9-16). In the real domain the first argument of R_C is smaller than the second for inverse circular functions and larger for logarithms and inverse hyperbolic functions, but in the complex domain the distinction disappears. In Section 6.10 we shall learn how to compute R_C numerically by successive quadratic transformations.

The inverse of (12) is

$$R_C(x, y) = R_{-1}[1, 1; x^{1/2} + (x - y)^{1/2}, x^{1/2} - (x - y)^{1/2}]. \quad (17)$$

Although the right side of (17) has the advantage of equal b parameters, the right side of (14) gains importance from being the degenerate case $y = z$ of the symmetric elliptic integral $R_{-1/2}(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}; x, y, z)$ (see Theorem 5.2-4 and Chapter 9).

We return now to (6.9-3) and its consequences for Bessel functions. From (5.8-23) we find

$$J_\mu(x) = \frac{(x/2)^\mu}{\Gamma(1 + \mu)} {}_0F_1(1 + \mu; -x^2/4). \quad (18)$$

This shows that every ${}_0F_1$ series is a Bessel function, as asserted in Section 2.4. The spherical Bessel function (5.8-24) is

$$j_L(x) = \frac{(x/2)^L}{(\frac{3}{2}, L)} {}_0F_1\left(\frac{3}{2} + L; -x^2/4\right), \quad L \in \mathbb{Z}. \quad (19)$$

By (2.4-14) and (2.4-13),

$$j_0(x) = \frac{\sin x}{x}, \quad j_{-1}(x) = \frac{\cos x}{x}. \quad (20)$$

The modified Bessel function usually denoted by I_μ is

$$I_\mu(x) = \frac{(x/2)^\mu}{\Gamma(1+\mu)} {}_0F_1(1+\mu; x^2/4) = \frac{(x/2)^\mu}{\Gamma(1+\mu)} S\left(\frac{1}{2} + \mu, \frac{1}{2} + \mu; x, -x\right). \quad (21)$$

Note that

$$J_\mu(ix) = i^\mu I_\mu(x). \quad (22)$$

By Exercise 6.9-19 or Corollary 6.3-7, $x^{-\mu}J_\mu(x)$ and $x^{-\mu}I_\mu(x)$ are entire functions of μ and x^2 .

Two simple properties of the ${}_0F_1$ series have important consequences for Bessel functions (see Exercises 6.9-20 and 6.9-21). The first is a differentiation formula,

$$\frac{d^n}{dx^n} \frac{{}_0F_1(c; x)}{\Gamma(c)} = \frac{{}_0F_1(c+n; x)}{\Gamma(c+n)}, \quad n \in \mathbb{N}, \quad (23)$$

and the second is a reflection formula,

$$\frac{{}_0F_1(1-n; x)}{\Gamma(1-n)} = x^n \frac{{}_0F_1(1+n; x)}{\Gamma(1+n)}, \quad n \in \mathbb{Z}. \quad (24)$$

The latter implies

$$J_{-n} = (-1)^n J_n, \quad I_{-n} = I_n, \quad n \in \mathbb{Z}. \quad (25)$$

We shall use (23) and (25) in Section 7.7.

6.10 The second quadratic transformation

The homogeneity parameter is halved by the first quadratic transformation (Section 6.9) but is left unchanged by the second quadratic transformation, which is a little harder to prove. Either alone or together with the first transformation, the second transformation gives a method of numerical computation for special cases of the R function. Moreover, it connects Gegenbauer polynomials with Jacobi polynomials.

TRANSFORMATION 6.10-1 Let $t \in \mathbb{C}$, $\beta + \frac{1}{2} \in U$, and $x, y \in \mathbb{C}_>$. Then

$$R_t(\beta, \beta; x^2, y^2) = R_t\left[2\beta + t, \frac{1}{2} - \beta - t; \left(\frac{x+y}{2}\right)^2, xy\right]. \quad (1)$$

Remark All four arguments lie in $\mathbb{C}_0$. Equation (1) is a special case of (9.5-1).

Proof Let $2\beta \in U$ and $|w|^{1/2} < \sqrt{2} - 1$, which implies $2|w|^{1/2} < 1 - |w| \leq |1 + w|$. By (6.3-6) and Theorem 6.9-1,

$$\begin{aligned}
 & R_t(\beta, \beta; 1 + w + 2\sqrt{w}, 1 + w - 2\sqrt{w}) \\
 &= (1 + w)^t \sum_{n=0}^{\infty} \frac{(-t, n)}{n!} R_n\left(\beta, \beta; \frac{-2\sqrt{w}}{1 + w}, \frac{2\sqrt{w}}{1 + w}\right) \\
 &= (1 + w)^t \sum_{m=0}^{\infty} \frac{(-t, 2m)(\frac{1}{2}, m)}{(2m)!(\beta + \frac{1}{2}, m)} \left(\frac{2\sqrt{w}}{1 + w}\right)^{2m} \\
 &= \sum_{m=0}^{\infty} \frac{(-t, 2m)}{(\beta + \frac{1}{2}, m)m!} w^m (1 + w)^{t-2m} \\
 &= \sum_{m=0}^{\infty} \sum_{k=0}^{\infty} \frac{(-t, 2m)(-t + 2m, k)}{(\beta + \frac{1}{2}, m)m!k!} (-1)^k w^{m+k}. \quad (2)
 \end{aligned}$$

The order of summation can be changed if the double series is absolutely convergent. We denote the general term by u_{mk} and define

$$A_m = \left| \frac{(-t, 2m)}{(\beta + \frac{1}{2}, m)m!} \right|.$$

Then

$$\begin{aligned}
 \sum_{m=0}^{\infty} \sum_{k=0}^{\infty} |u_{mk}| &\leq \sum_{m=0}^{\infty} \sum_{k=0}^{\infty} A_m \frac{(|t| + 2m, k)}{k!} |w|^{m+k} \\
 &= \sum_{m=0}^{\infty} A_m |w|^m (1 - |w|)^{-|t| - 2m}.
 \end{aligned}$$

This series converges by the ratio test since $A_{m+1}/A_m \rightarrow 4$ as $m \rightarrow \infty$ and $4|w| < (1 - |w|)^2$. Therefore the double series converges absolutely.

In the last member of (2) we set $k = n - m$ and note that

$$\begin{aligned}
 \frac{(-t, 2m)(-t + 2m, k)(-1)^k}{(\beta + \frac{1}{2}, m)} &= \frac{(-t, n + m)(-1)^{n-m}}{(\beta + \frac{1}{2}, n)(\beta + \frac{1}{2} + n, m - n)} \\
 &= \frac{(-t, n)(-t + n, m)(\frac{1}{2} - \beta - n, n - m)}{(\beta + \frac{1}{2}, n)}.
 \end{aligned}$$

Applying (2.3-1) and (6.2-5), we find

$$\begin{aligned}
 & R_t[\beta, \beta; (1 + \sqrt{w})^2, (1 - \sqrt{w})^2] \\
 &= \sum_{n=0}^{\infty} \frac{(-t, n)}{(\beta + \frac{1}{2}, n)} w^n \sum_{m=0}^n \frac{(-t + n, m)(\frac{1}{2} - \beta - n, n - m)}{m!(n - m)!} \\
 &= \sum_{n=0}^{\infty} \frac{(-t, n)(\frac{1}{2} - \beta - t, n)}{(\beta + \frac{1}{2}, n)n!} w^n \\
 &= \sum_{n=0}^{\infty} \frac{(-t, n)}{n!} R_n\left(\frac{1}{2} - \beta - t, 2\beta + t; w, 0\right) \\
 &= R_t\left(\frac{1}{2} - \beta - t, 2\beta + t; 1 - w, 1\right). \tag{3}
 \end{aligned}$$

Now we assume $|x - y| < (\sqrt{2} - 1)|x + y|$ and set $w = (x - y)^2/(x + y)^2$. Using the homogeneity and symmetry of R , we obtain (1). By Theorem 6.8-2 both sides of (1) are holomorphic in (x, y) on $\mathbb{C}_{>}^2$, and so the equality holds on $\mathbb{C}_{>}^2$ by the permanence of functional relations. By Theorem 6.8-4 the assumption $2\beta \in U$ can be relaxed to $\beta + \frac{1}{2} \in U$. $\square$

EXAMPLE 6.10-2 (Gauss's arithmetic-geometric mean) If $-t = \beta = \frac{1}{2}$ the parameters on the two sides of (1) are the same. Defining

$$R_K(\zeta, \eta) = R_{-1/2}(\frac{1}{2}, \frac{1}{2}; \zeta, \eta), \tag{4}$$

we find

$$R_K(x^2, y^2) = R_K\left[\left(\frac{x + y}{2}\right)^2, xy\right], \quad x, y \in \mathbb{C}_{>}. \tag{5}$$

Thus R_K is invariant under a quadratic transformation which replaces x and y by their arithmetic and geometric means, where $(xy)^{1/2} \in \mathbb{C}_{>}$. If $(x - y)/(x + y) = \varepsilon$ and $|\varepsilon| \ll 1$, the relative difference of the means is of order ε^2 . Successive applications of (5) cause the arguments of R_K to agree to a number of significant figures that doubles with each iteration. The common limit of the two sequences defined by

$$\begin{aligned}
 x_0 &= x, & y_0 &= y, \\
 x_{n+1} &= \frac{1}{2}(x_n + y_n), & y_{n+1} &= (x_n y_n)^{1/2}, \quad n \in \mathbb{N},
 \end{aligned} \tag{6}$$

is known as Gauss's arithmetic-geometric mean of x and y ,

$$M = M(x, y) = \lim x_n = \lim y_n. \tag{7}$$

Since R_K is a continuous function of its arguments,

$$R_K(x^2, y^2) = R_K(x_n^2, y_n^2) = \lim_{n \rightarrow \infty} R_K(x_n^2, y_n^2) = R_K(M^2, M^2) = \frac{1}{M(x, y)}. \tag{8}$$

The last equality is a case of (5.2-2). Computation of M by (6) and (7) produces the value of R_K to any prescribed accuracy.

This algorithm led Gauss in 1799 to the discovery of elliptic functions. He began with (6) and found (first in the case $x = 1$, $y = \sqrt{2}$) that the reciprocal of $M(x, y)$ is represented by

$$\frac{2}{\pi} \int_0^{\pi/2} (x^2 \sin^2 \theta + y^2 \cos^2 \theta)^{-1/2} d\theta = R_K(x^2, y^2). \quad (9)$$

The equality follows from setting $\sin^2 \theta = u$ and using (5.9-1). The integral is a symmetrical variant of Legendre's complete elliptic integral of the first kind,

$$K(k) = \int_0^{\pi/2} (1 - k^2 \sin^2 \theta)^{-1/2} d\theta = \frac{\pi}{2} R_K(1 - k^2, 1), \quad (10)$$

which is seen by writing $1 - k^2 \sin^2 \theta = (1 - k^2) \sin^2 \theta + \cos^2 \theta$. The symmetry of R_K in its arguments is equivalent to a linear transformation of K (Exercise 6.10-5), and the invariance property (5) is equivalent to the Landen transformation of K (Exercise 6.10-5).

TRANSFORMATION 6.10-3 Let $n \in \mathbb{N}$ and $\beta, x, y \in \mathbb{C}$. Then

$$(2\beta, n)R_n(\beta, \beta; x^2, y^2) = (\beta, n)R_n[\tfrac{1}{2} - \beta - n, \tfrac{1}{2} - \beta - n; (x + y)^2, (x - y)^2]. \quad (11)$$

Remark This is the only transformation of one R function into another in which both sides have equal and variable b parameters. It is its own inverse.

Proof Since the left side is a polynomial in β, x, y , it suffices to prove (11) when $2\beta \in U$ and $x, y \in \mathbb{C}_>$. Set $t = n$ in (1) and apply the linear transformation (6.5-1) to the right side. $\square$

For an R polynomial in two variables with initially equal b parameters, the right side of (1) or (11) provides a third starting point (see Section 6.9) for linear transformations, leading to six more choices of arguments. We now have eighteen choices, each corresponding to a different argument of an equivalent ${}_2F_1$ polynomial. For example, setting $x = e^{i\theta/2}$ and $y = e^{-i\theta/2}$ in (11), we find that a Gegenbauer polynomial with argument $\cos \theta$ can be written as a ${}_2F_1$ polynomial with argument $1 - (\cos^2 \frac{1}{2}\theta)/(-\sin^2 \frac{1}{2}\theta) = \csc^2 \frac{1}{2}\theta$. From (1) we get the argument $1 - \cos^2 \frac{1}{2}\theta = \sin^2 \frac{1}{2}\theta$. Other choices are $\sec^2 \frac{1}{2}\theta$, $\cos^2 \frac{1}{2}\theta$, $-\tan^2 \frac{1}{2}\theta$, and $-\cot^2 \frac{1}{2}\theta$.

Eighteen choices are too many, and we avoid this embarrassment of riches by giving preference always to R polynomials with equal b parameters. Symmetry implies that the order of the arguments is immaterial, and

thus the eighteen choices are reduced to two, which are connected by (11). By (6.7-21),

$$\begin{aligned} \frac{C_n^v(\cos \theta)}{C_n^v(1)} &= R_n(v, v; e^{i\theta}, e^{-i\theta}) \\ &= \frac{2^n(v, n)}{(2v, n)} R_n\left(\frac{1}{2} - v - n, \frac{1}{2} - v - n; \cos \theta + 1, \cos \theta - 1\right), \end{aligned} \quad (12)$$

where $C_n^v(1) = (2v, n)/n!$. Because of the form of the third member it will be easy in Chapter 7 to identify Gegenbauer polynomials as Jacobi polynomials with equal b parameters. In particular, the case $v = \frac{1}{2}$ is the Legendre polynomial,

$$P_n(\cos \theta) = R_n\left(\frac{1}{2}, \frac{1}{2}; e^{i\theta}, e^{-i\theta}\right) = \frac{2^n(\frac{1}{2}, n)}{n!} R_n(-n, -n; \cos \theta + 1, \cos \theta - 1). \quad (13)$$

The Chebyshev polynomial is

$$\begin{aligned} T_n(\cos \theta) &= \cos n\theta = \lim_{v \rightarrow 0} R_n(v, v; e^{i\theta}, e^{-i\theta}) \\ &= (1 + \delta_{n0})2^{n-1} R_n\left(\frac{1}{2} - n, \frac{1}{2} - n; \cos \theta + 1, \cos \theta - 1\right), \end{aligned} \quad (14)$$

where δ_{n0} is the Kronecker symbol (with value 1 if $n = 0$ and 0 otherwise). As we shall see in Section 7.7, the appearance of the Kronecker delta here is related to its appearance in the customary form of the Fourier series,

$$\frac{1}{2} a_0 + \sum_{n=1}^{\infty} a_n \cos n\theta = \sum_{n=0}^{\infty} \frac{a_n \cos n\theta}{1 + \delta_{n0}}. \quad (15)$$

Instead of normalizing the Chebyshev polynomial so that $T_n(1) = 1$, as in (14), it is often better to use the polynomial

$$R_n\left(\frac{1}{2} - n, \frac{1}{2} - n; x + 1, x - 1\right), \quad n \in \mathbb{N}. \quad (16)$$

This is called a monic polynomial of degree n because the coefficient of x^n is unity, as one sees by (6.4-2) or alternatively by taking the product with x^{-n} , using homogeneity, and letting $x \rightarrow \infty$. Chebyshev discovered that the polynomial (16) has a property which characterizes it uniquely: its maximum absolute value on the interval $[-1, 1]$ is smaller than that of any other monic polynomial of degree n . This minimax property accounts for its usefulness in numerical analysis because it means that expansions in Chebyshev polynomials tend to converge rapidly. (Except in contrived

examples the expansion coefficients seldom change drastically in magnitude if monic Chebyshev polynomials are replaced by other monic polynomials, for example monic Gegenbauer polynomials of order $\nu \neq 0$.) To approximate a given function on $[-1, 1]$ by a polynomial of degree m , one expands the function (see Chapter 7) in an infinite series of polynomials (16) and discards terms of degree higher than m . The truncated series approximates the given function with a maximum error that is usually close to the smallest maximum error for any polynomial of degree m . It is almost always smaller than that for least-squares approximation, which minimizes an average error but often has a large maximum error. [For theory and applications of Chebyshev polynomials, see Fox (1968), Rivlin (1974), and Luke (1975).]

The associated Legendre function $P_n^m(x)$ is often defined for $m, n \in \mathbb{N}$ and $-1 \leq x \leq 1$ by

$$P_n^m(x) = (-1)^m (1-x^2)^{m/2} (d/dx)^m P_n(x). \quad (17)$$

In quantum theory the factor $(-1)^m$ is called the Condon–Shortley choice of phase; it has the effects that $P_1^1(\cos \theta) = -\sin \theta$ and the matrix elements of angular momentum operators are nonnegative. Using (13), Exercise 5.9-18, and (5.9-21),

$$\begin{aligned} P_n^m(x) &= (-1)^m \frac{2^n(\frac{1}{2}, n)}{(n-m)!} (1-x^2)^{m/2} R_{n-m}(-n, -n; x+1, x-1) \\ &= \frac{2^n(\frac{1}{2}, n)}{(n-m)!} (1-x^2)^{-m/2} R_{n+m}(-n, -n; x+1, x-1). \end{aligned} \quad (18)$$

The last member serves to define $P_n^m(x)$ for $-1 \leq x \leq 1$ if $n \in \mathbb{N}$ and $m = -1, -2, \dots, -n$. Comparison with the middle member shows that

$$\frac{P_n^{-m}}{(n-m)!} = (-1)^m \frac{P_n^m}{(n+m)!}, \quad m = 0, \pm 1, \dots, \pm n. \quad (19)$$

By (18) and (12), P_n^m can be expressed in terms of the Gegenbauer polynomial $C_{n-m}^{1/2+m}$ or $C_{n+m}^{1/2-m}$. Application of the quadratic transformation (11) to the last member of (18) allows P_n^m as defined here for $-1 \leq x \leq 1$ to be compared with the function defined by (6.8-19) on a cut plane. Note that the latter function is entire in μ by Theorem 6.8-4 and so is defined when μ is an integer. The only formal difference between the two functions is that $(1-x)^{-m/2}$ on the cut is replaced by $(x-1)^{-m/2}$ off the cut.

There are only two independent quadratic transformations of the R function in two variables, but many hybrids can be produced by combining

them with each other and with the transformations (5.9-19)–(5.9-21). One such hybrid is especially important because it leads to an invariance property of R_C which can be used for numerical computation.

TRANSFORMATION 6.10-4 Let $t \in \mathbb{C}$, $\frac{1}{2} - t + \beta \in U$, and $x, y \in \mathbb{C}_>$. Then

$$R_t\left(\beta, \frac{1}{2} - t; x^2, y^2\right) = R_{2t}\left(2\beta, \frac{1}{2} - t - \beta; \frac{x+y}{2}, y\right). \quad (20)$$

Proof Eliminate the R function with equal b parameters between (6.9-7) and (1) to obtain

$$R_t\left[\beta + t, \frac{1}{2} - t; \left(\frac{r^2 + s^2}{2}\right)^2, r^2 s^2\right] = R_{2t}\left[2\beta + 2t, \frac{1}{2} - \beta - 2t; \left(\frac{r+s}{2}\right)^2, rs\right]. \quad (21)$$

Replace β by $\beta - t$ and set $\frac{1}{2}(r \pm s) = [\frac{1}{2}(x \pm y)]^{1/2}$. $\square$

EXAMPLE 6.10-5 (*Borchardt's algorithm*) Logarithms, inverse circular functions, and inverse hyperbolic functions can be computed numerically by an algorithm first suggested by Gauss in 1800 but rediscovered and published by Borchardt in 1880. Set $-t = \beta = \frac{1}{2}$ in (20) and use (6.9-14); thus

$$R_C(x^2, y^2) = R_{-1}\left(1, \frac{1}{2}; \frac{x+y}{2}, y\right). \quad (22)$$

By (5.9-20) this becomes an invariance property,

$$R_C(x^2, y^2) = R_C\left[\left(\frac{x+y}{2}\right)^2, \frac{x+y}{2}y\right], \quad x, y \in \mathbb{C}_>, \quad (23)$$

which remains valid by continuity if $x = 0$.

To arrange (23) for successive applications we set

$$x^2 = x_n^2, \quad y^2 = y_n^2, \quad \left(\frac{x+y}{2}\right)^2 = x_{n+1}^2, \quad \frac{x+y}{2}y = y_{n+1}^2.$$

Then $R_C(x_n^2, y_n^2)$ is independent of n if

$$x_{n+1} = \frac{1}{2}(x_n + y_n), \quad y_{n+1} = (x_{n+1}y_n)^{1/2}, \quad (24)$$

the sign of the square root being taken so that $y_{n+1} \in \mathbb{C}_>$. Since (24) implies

$$x_n^2 - y_n^2 = 4^{-n}(x_0^2 - y_0^2), \quad (25)$$

x_n and y_n have a common limit L as $n \rightarrow \infty$. By continuity of R_C ,

$$R_C(x_n^2, y_n^2) \rightarrow R_C(L^2, L^2) = 1/L.$$

In summary, if $x_0 \in \mathbb{C}_> \cup \{0\}$ and $y_0 \in \mathbb{C}_>$, define x_n and y_n by (24) for every $n \in \mathbb{N}$ and let L be their common limit as $n \rightarrow \infty$. Then

$$R_C(x_0^2, y_0^2) = \frac{1}{L}. \quad (26)$$

To compute a logarithm, inverse circular function, or inverse hyperbolic function, we express it in terms of R_C by (6.9-15) or (6.9-16) and calculate L by successive applications of (24). According to (25) the error committed in approximating L by x_n or y_n ultimately decreases by a factor of four in each cycle of iteration. A factor of 16 per cycle is achieved if L is approximated by

$$\sigma_n = \frac{1}{3}(x_n + 2y_n), \quad (27)$$

because (25) and Exercise 6.9-18 imply

$$\frac{\sigma_n}{L} = 1 + 4^{-2n-1}(x_0^2 - y_0^2)^2/45L^4 + O(4^{-3n}). \quad (28)$$

By an extrapolation method it can be shown that the factor of 16 is improved to 64 if L is approximated by

$$\sigma_n + \frac{1}{15}(\sigma_n - \sigma_{n-1}), \quad (29)$$

and 64 is improved to 256 by

$$\sigma_n + \left(\frac{1}{15} + \frac{1}{63}\right)(\sigma_n - \sigma_{n-1}) + \frac{1}{15 \cdot 63}(\sigma_n - 2\sigma_{n-1} + \sigma_{n-2}). \quad (30)$$

Further improvements are given by Carlson (1972b) in a form convenient for programmed computation.

In Section 9.6 we shall see that (23) is a degenerate case of a symmetric duplication theorem for an elliptic integral.

6.11 Gegenbauer's product formula

In a generating relation such as (6.6-1),

$$\prod_{i=1}^k (1 - tz_i)^{-b_i} = \sum_{n=0}^{\infty} t^n \frac{(c, n)}{n!} R_n(b, z), \quad |tz| < 1, \quad (1)$$

the general term of the series is a multiple of an R polynomial. In a two-sided or bilateral generating relation, the general term is a product of two polynomials. We shall now prove such a generating relation for products of R polynomials and specialize it to products of Gegenbauer polynomials. It

will then be a short step to Gegenbauer's integral representation for a product of Gegenbauer polynomials. In Section 7.3 we shall use this important representation to prove the addition theorem for Gegenbauer polynomials.

GENERATING RELATION 6.11-1 Let $z \in \mathbb{C}^k$, $b \in U_k$, $c = \sum_{i=1}^k b_i$, and $a, x, y \in \mathbb{C}$. Assume $|xz| < 1$ and $|yz| < 1$, and let $(1 - xz)/(1 - yz)$ denote the k -tuple with i th component $(1 - xz_i)/(1 - yz_i)$. Then

$$\prod_{i=1}^k (1 - yz_i)^{-b_i} R_{-a} \left(b, \frac{1 - xz}{1 - yz} \right) = \sum_{n=0}^{\infty} \frac{(c, n)}{n!} R_n(a, c - a; x, y) R_n(b, z). \quad (2)$$

Remark This reduces to (1) if $x = y$ and to (6.6-4) if $y = 0$. The use of double Dirichlet averages [Carlson (1971b)] shows the direct relation of (2) to the binomial series.

Proof If we set $t = ux + (1 - u)y$ in (1), the series converges uniformly on $0 \leq u \leq 1$. Assume for the moment that $\operatorname{Re} c > \operatorname{Re} a > 0$ and define $\alpha = (a, c - a)$. Integrating term by term with respect to the Dirichlet measure μ_α , we have

$$\begin{aligned} & \int_0^1 \prod_{i=1}^k [u(1 - xz_i) + (1 - u)(1 - yz_i)]^{-b_i} d\mu_\alpha(u) \\ &= \sum_{n=0}^{\infty} \frac{(c, n)}{n!} R_n(b, z) \int_0^1 [ux + (1 - u)y]^n d\mu_\alpha(u). \end{aligned}$$

By (5.7-1) and (6.8-2) this is the same as (2), since $(1 - xz)/(1 - yz) \in \mathbb{C}_0^k$.

To remove the conditions on the real parts of a and c , we divide (2) by $\Gamma(c)$, making the left side and every term on the right side entire in a and b by Theorems 6.8-2 and 6.2-2. In order to show that the sum of the series is entire, we choose $r > 0$ and assume $|a| \leq r$ and $\|b\| \leq r$, whence $|a| + |c - a| \leq 3r$. By (6.2-24) and (6.2-27) the series (except for terms with $n < r + 2$) is majorized by

$$e^{\pi r/2} \sum_n \frac{(3r, n)(r, n)}{n! \Gamma(n - r)} \xi^n,$$

where $\xi = \max\{|xz|, |yz|\} < 1$. This series converges by the ratio test. Hence the original series converges uniformly by the Weierstrass M test, and its sum is holomorphic in a and b for $|a| \leq r$ and $\|b\| \leq r$, where r is arbitrarily large. After division by $\Gamma(c)$, (2) holds for all complex a and b by the permanence of functional relations. We can then multiply by $\Gamma(c)$ if $b \in U_k$. $\square$

COROLLARY 6.11-2 (Meixner's formula) Let $a, \beta, x, y, X, Y \in \mathbb{C}$ and $c \in U$. Assume xX, xY, yX , and yY have modulus strictly less than 1. Then

$$\begin{aligned} & (1 - yX)^{a-\beta}(1 - yY)^{a+\beta-c} \\ & \quad \times R_{-a}[\beta, c - \beta; (1 - xX)(1 - yY), (1 - xY)(1 - yX)] \\ & = \sum_{n=0}^{\infty} \frac{(c, n)}{n!} R_n(a, c - a; x, y) R_n(\beta, c - \beta; X, Y). \end{aligned} \quad (3)$$

Proof Set $k = 2$, $b = (\beta, c - \beta)$, $z = (X, Y)$ in (2), and use the homogeneity of R_{-a} . $\square$

COROLLARY 6.11-3 (Ossicini's formula) Let $2v \in U$ and $t, \theta, \varphi \in \mathbb{C}$. Assume $|t| < \exp(-|\operatorname{Im} \theta| - |\operatorname{Im} \varphi|)$. If C_n^v denotes Gegenbauer's polynomial, then

$$\begin{aligned} & R_{-v}[v, v; 1 - 2t \cos(\theta + \varphi) + t^2, 1 - 2t \cos(\theta - \varphi) + t^2] \\ & = \sum_{n=0}^{\infty} t^n C_n^v(\cos \theta) C_n^v(\cos \varphi) / C_n^v(1). \end{aligned} \quad (4)$$

Proof In (3) set $a = \beta = v$, $c = 2v$, $x = te^{i\theta}$, $y = te^{-i\theta}$, $X = e^{i\varphi}$, $Y = e^{-i\varphi}$, and use (6.7-21). $\square$

Equation (3), expressed in terms of ${}_2F_1$ functions, was discovered by Meixner (1942) and rediscovered by Weisner (1955) by a group-theoretical method. Equation (4), found by Ossicini (1952) by still another method, is a special case not only of Meixner's formula but also of a bilateral expansion in Jacobi polynomials due to Bailey (1938).

THEOREM 6.11-4 (Gegenbauer's product formula) Let $n \in \mathbb{N}$, $v \in \mathbb{C}_+$, and $\theta, \varphi \in \mathbb{C}$. Let $\mu_{(v, v)}$ denote the Dirichlet measure μ_b with $b = (v, v)$. Then

$$\begin{aligned} & C_n^v(\cos \theta) C_n^v(\cos \varphi) \\ & = C_n^v(1) \int_0^1 C_n^v[u \cos(\theta + \varphi) + (1 - u) \cos(\theta - \varphi)] d\mu_{(v, v)}(u). \end{aligned} \quad (5)$$

Remark If we define $f(x) = C_n^v(x)/C_n^v(1)$, then (5) states that $f(\cos \theta) f(\cos \varphi) = F[v, v; \cos(\theta + \varphi), \cos(\theta - \varphi)]$. In this form it holds for $v + \frac{1}{2} \in U$, as one can show by using Theorem 6.2-6, Corollary 6.3-7, and the permanence of functional relations.

Proof A convex combination of the arguments on the left side of (4) is

$$\begin{aligned} & u[1 - 2t \cos(\theta + \varphi) + t^2] + (1 - u)[1 - 2t \cos(\theta - \varphi) + t^2] \\ & = 1 - 2tx(u) + t^2, \end{aligned}$$

where

$$x(u) = u \cos(\theta + \varphi) + (1 - u) \cos(\theta - \varphi).$$

By (5.2-1) and (6.7-19), if $|t|$ is sufficiently small,

$$\begin{aligned} R_{-,v}[v, v; 1 - 2t \cos(\theta + \varphi) + t^2, 1 - 2t \cos(\theta - \varphi) + t^2] \\ = \int_0^1 [1 - 2tx(u) + t^2]^{-v} d\mu_{(v, v)}(u) \\ = \int_0^1 \sum_{n=0}^{\infty} t^n C_n^v[x(u)] d\mu_{(v, v)}(u). \end{aligned}$$

We may integrate the series term by term as a formal power series (see Section 2.3) since we wish only to pick out the coefficient of t^n . Comparison with the same coefficient in (4) proves (5). $\square$

The substitution $u = \sin^2(\psi/2)$ expresses (5) in a form widely used. The case $v = \frac{1}{2}$ becomes

$$\frac{1}{\pi} \int_0^\pi P_n(\cos \theta \cos \varphi + \sin \theta \sin \varphi \cos \psi) d\psi = P_n(\cos \theta) P_n(\cos \varphi). \quad (6)$$

If two directions in three-dimensional Euclidean space have polar angles θ and φ , and if their azimuthal angles differ by ψ , the cosine of the angle between them is $\cos \theta \cos \varphi + \sin \theta \sin \varphi \cos \psi$. In physical problems with axial symmetry an integration over azimuthal angles often leads to the left side of (6).

NOTES

Rainville (1960) treats special polynomials in considerable detail. Boas (1964) discusses Appell sets of polynomials and their generalizations, and Scaravelli (1965) gives a bibliography. Carlson (1970b) considers Appell sets in several variables. For a popular account of Fibonacci numbers see Gardner (1969) as well as the references in Section 6.6; these numbers are discussed in complete detail in various journal articles in the *Fibonacci Quarterly*. For Gegenbauer polynomials and Gegenbauer functions see Durand (1976), Robin (1957, Chapter X), Szegő (1975), Erdélyi (1953, Section 3.15, Chapters X and XI), and Rainville (1960). For Legendre functions see MacRobert (1967), Erdélyi (1953, Chapter III), Robin (1957), Hobson (1931), and Whittaker (1927, Chapter XV). For Bessel functions see Watson (1944), Whittaker (1927, Chapter XVII), Erdélyi (1953, Chapter VII), Abramowitz (1964, Chapters 9-11), Petiau (1955), and Luke (1962).

If t is real while c and the w_i and z_i are positive, then $[R_i(cw, z)]^{1/i}$ is a homogeneous mean value of the z_i which increases monotonically with t (see the Remark following the proof of Theorem 6.2-5). For properties of this mean value and its use in finding inequalities for hypergeometric functions, see Carlson (1965b, 1966, 1968) and Tobey (1967).

A class of algorithms including those of Gauss and Borchardt is discussed by Carlson (1971a).

FORMULARY

$$R_n(b, z) = \frac{n!}{(c, n)} \sum \frac{(b_1, m_1) \cdots (b_k, m_k)}{m_1! \cdots m_k!} z_1^{m_1} \cdots z_k^{m_k}, \quad c = \sum_{i=1}^k b_i, \quad (6.2-1)$$

$$R_n(\beta, \beta'; x, y) = \frac{n!}{(\beta + \beta', n)} \sum_{m=0}^n \frac{(\beta, m)(\beta', n-m)}{m!(n-m)!} x^m y^{n-m}, \quad (6.2-3)$$

$$(c_k, n)R_n(b_1, \dots, b_k; z_1, \dots, z_p, 0, \dots, 0) \\ = (c_p, n)R_n(b_1, \dots, b_p; z_1, \dots, z_p), \quad c_r = \sum_{i=1}^r b_i, \quad (6.2-6)$$

$$C_n(z) = \frac{(k, n)}{n!} R_n(1, \dots, 1; z_1, \dots, z_k), \quad n \in \mathbb{N}, \quad (6.2-11)$$

$$E_n(z) = \binom{k}{n} R_n(-1, \dots, -1; z_1, \dots, z_k), \quad n = 0, 1, \dots, k, \quad (6.2-12)$$

$$\lim_{c \rightarrow 0} R_n(cw, z) = \sum_{i=1}^k w_i z_i^n, \quad \sum_{i=1}^k w_i = 1, \quad (6.2-17)$$

$$\lim_{c \rightarrow \infty} R_n(cw, z) = \left(\sum_{i=1}^k w_i z_i \right)^n, \quad \sum_{i=1}^k w_i = 1, \quad (6.2-18)$$

$$|(c, n)R_n(b, z)| \leq (\|b\|, n)|z|^n, \quad |z| = \max_i |z_i|, \quad (6.2-24)$$

$$|(c, n)R_n(b, z)| \leq (\|b\|, n)\|z\|^n, \quad \|z\| = \sum_{i=1}^k |z_i|, \quad (6.2-25)$$

$$F(b_1, \dots, b_p, 0, \dots, 0; z_1, \dots, z_k) = F(b_1, \dots, b_p; z_1, \dots, z_p), \quad (6.3-3)$$

$$R_n(b, z + \lambda) = \sum_{m=0}^n \binom{n}{m} \lambda^{n-m} R_m(b, z), \quad (6.4-2)$$

$$(c, n)R_n(b, z) = (b_1, n)R_n(1 - c - n, b_2, \dots, b_k; z_1, z_1 - z_2, \dots, z_1 - z_k), \quad (6.5-3)$$

$$\prod_{i=1}^k (1 - tz_i)^{-b_i} = \sum_{n=0}^{\infty} t^n \frac{(c, n)}{n!} R_n(b, z), \quad (6.6-1)$$

$$R_{-c}(b, z) = \prod_{i=1}^k z_i^{-b_i} = \int_E (u \cdot z)^{-c} d\mu_b(u), \quad c = \sum_{i=1}^k b_i, \quad (6.6-5)$$

$$\Delta^n f(x) = F^{(n)}(1, \dots, 1; x, x+1, \dots, x+n), \quad (6.6-11)$$

$$S_n^{(m)} = (-1)^{n-m} \binom{n}{m} R_{n-m}(-1, \dots, -1; 0, 1, \dots, n-1), \quad (6.6-15)$$

$$\mathfrak{S}_n^{(m)} = \binom{n}{m} R_{n-m}(1, \dots, 1; 0, 1, \dots, m), \quad (6.6-17)$$

$$F_n = nR_{n-1} \left(1, 1; \frac{1+\sqrt{5}}{2}, \frac{1-\sqrt{5}}{2} \right), \quad (6.6-22)$$

$$P_n(\cos \theta) = R_n\left(\frac{1}{2}, \frac{1}{2}; e^{i\theta}, e^{-i\theta}\right), \quad (6.7-5)$$

$$C_n^v(\cos \theta) = \frac{(2v, n)}{n!} R_n(v, v; e^{i\theta}, e^{-i\theta}), \quad (6.7-21)$$

$$T_n(\cos \theta) = \cos n\theta = \lim_{v \rightarrow 0} R_n(v, v; e^{i\theta}, e^{-i\theta}), \quad (6.7-30)$$

$$R_{-a}(b, z) = \frac{1}{2\pi i} \int_{\gamma} R_{-1}(a, c-a; s-1, s) \prod_{i=1}^k (1-s+sz_i)^{-b_i} ds, \quad (6.8-5)$$

$$\frac{1}{\Gamma(2v)} R_x(v, v; x, y) = \frac{\Gamma(1+\alpha)}{\Gamma(2v+\alpha)} (xy)^{\alpha/2} C_\alpha^v \left[\frac{x+y}{2(xy)^{1/2}} \right], \quad (6.8-9)$$

$$R_{2n+1}(\beta, \beta; x, -x) = 0, \quad (6.9-1)$$

$$R_{2n}(\beta, \beta; x, -x) = \frac{(\frac{1}{2}, n)}{(\beta + \frac{1}{2}, n)} x^{2n}, \quad (6.9-2)$$

$$S(\beta, \beta; x, y) = e^{(x+y)/2} {}_0F_1[\beta + \frac{1}{2}; (x-y)^2/16], \quad (6.9-3)$$

$$R_{2t}(\beta, \beta; x, y) = R_t \left[\beta + t, \frac{1}{2} - t; \left(\frac{x+y}{2} \right)^2, xy \right], \quad (6.9-7)$$

$$\arcsin x = x R_C(1-x^2, 1), \quad \log x = (x-1) R_C \left[\left(\frac{x+1}{2} \right)^2, x \right]. \quad (6.9-15, 16)$$

$$J_\mu(x) = \frac{(x/2)^\mu}{\Gamma(1+\mu)} {}_0F_1(1+\mu; -x^2/4), \quad (6.9-18)$$

$$R_t(\beta, \beta; x^2, y^2) = R_t \left[2\beta + t, \frac{1}{2} - \beta - t; \left(\frac{x+y}{2} \right)^2, xy \right], \quad (6.10-1)$$

$$(2\beta, n) R_n(\beta, \beta; x^2, y^2) = (\beta, n) R_n \left[\frac{1}{2} - \beta - n, \frac{1}{2} - \beta - n; (x+y)^2, (x-y)^2 \right], \quad (6.10-11)$$

$$C_n^v(\cos \theta) = \frac{2^n(v, n)}{n!} R_n \left(\frac{1}{2} - v - n, \frac{1}{2} - v - n; \cos \theta + 1, \cos \theta - 1 \right), \quad (6.10-12)$$

$$T_n(\cos \theta) = \cos n\theta = (1 + \delta_{n0}) 2^{n-1} R_n \left(\frac{1}{2} - n, \frac{1}{2} - n; \cos \theta + 1, \cos \theta - 1 \right), \quad (6.10-14)$$

$$P_n^m(x) = (-1)^m \frac{2^n(\frac{1}{2}, n)}{(n-m)!} (1-x^2)^{m/2} R_{n-m}(-n, -n; x+1, x-1), \quad (6.10-18)$$

$$\prod_{i=1}^k (1-yz_i)^{-b_i} R_{-a} \left(b, \frac{1-xz}{1-yz} \right) = \sum_{n=0}^{\infty} \frac{(c, n)}{n!} R_n(a, c-a; x, y) R_n(b, z), \quad (6.11-2)$$

$$\begin{aligned} R_{-v}[v, v; 1-2t \cos(\theta + \varphi) + t^2, 1-2t \cos(\theta - \varphi) + t^2] \\ = \sum_{n=0}^{\infty} t^n C_n^v(\cos \theta) C_n^v(\cos \varphi) / C_n^v(1), \end{aligned} \quad (6.11-4)$$

$$C_n^v(\cos \theta) C_n^v(\cos \varphi) = C_n^v(1) \int_0^1 C_n^v[u \cos(\theta + \varphi) + (1-u) \cos(\theta - \varphi)] d\mu_{(v, v)}(u). \quad (6.11-5)$$

EXERCISES

6.2-1 If $\alpha, x \in \mathbb{C}$, show that

$$\sum_{m=0}^n \frac{(\alpha, m)}{m!} x^m = \frac{(\alpha+1, n)}{n!} R_n(\alpha, 1; x, 1).$$

(This polynomial is a truncated binomial series. The remainder of the series is expressed as an R function by Exercise 5.9-1.)

6.2-2 Let $b, z \in \mathbb{C}^k$ and $c = \sum_{i=1}^k b_i$. If $\beta, \zeta \in \mathbb{C}$, let b, β stand for the array $b_1, \dots, b_k, \beta$, and similarly for z, ζ . Show that

$$\frac{(c + \beta, n)}{n!} R_n(b, \beta; z, \zeta) = \sum_{m=0}^n \frac{(c, m)(\beta, n-m)}{m!(n-m)!} R_m(b, z) \zeta^{n-m}.$$

6.2-3 It is easy to show from (6.2-9) and (6.2-12) that

$$\begin{aligned} E_n(z_1, \dots, z_k) &= E_n(z_1, \dots, z_{k-1}) + z_k E_{n-1}(z_1, \dots, z_{k-1}), \quad n \leq k-1, \\ C_n(z_1, \dots, z_k) &= C_n(z_1, \dots, z_{k-1}) + z_k C_{n-1}(z_1, \dots, z_k). \end{aligned}$$

Deduce these results from relations between associated R functions.

6.2-4 Let $n \in \mathbb{N}$ and $x, y \in \mathbb{C}$. Show that

$$R_n(1, 1; x, y) = (n+1)^{-1} (x^n + x^{n-1}y + \dots + xy^{n-1} + y^n).$$

6.2-5 Let $n \in \mathbb{N}$ and $x, y, \beta, \gamma \in \mathbb{C}$. Assume $y \neq 0$ and $(\gamma, n) \neq 0$. Show that

$$R_n(\beta, \gamma - \beta; x, y) = y^n {}_2F_1(-n, \beta; \gamma; 1 - x/y).$$

6.2-6 Let $n \in \mathbb{N}$ and $b, z \in \mathbb{C}^k$. Define $c = \sum_{i=1}^k b_i$ and $\|b\| = \sum_{i=1}^k |b_i|$. As $\|b\| \rightarrow 0$ show that

$$R_n(b, z)/\Gamma(c) \sim \sum_{i=1}^k b_i z_i^n \rightarrow 0.$$

6.2-7 Let $n \in \mathbb{N}$ and $b, z \in \mathbb{C}^k$. Define $c = \sum_{i=1}^k b_i$. Show that

$$\limsup_{n \rightarrow \infty} \left| \frac{R_n(b, z)}{\Gamma(c)} \right|^{1/n} \leq \|z\|.$$

The proof of Theorem 6.6-2 shows that equality holds if $b \in U^k$.

6.2-8 Let $n \in \mathbb{N}$, $c \in \mathbb{C}$, and $z, w \in \mathbb{C}^k$. Assume $\sum_{i=1}^k w_i = 1$ and $0 < |c| \leq \frac{1}{2}$. Show that

$$|R_n(cw, z)| \leq \frac{(\frac{1}{2}\|w\|, n)}{(-\frac{1}{2}, n)} |z|^n.$$

6.2-9 Let $n \in \mathbb{N}$ and $x, y \in \mathbb{C}$. Show that

$$R_n(1, -n; x, y)/\Gamma(1-n) = n!(x-y)^n.$$

6.2-10 Let $n \in \mathbb{N}$, $z \in \mathbb{C}^k$, and $b = (\beta, \beta, \dots, \beta) \in \mathbb{C}^k$. Let $s-1$ be the largest integer not exceeding $(n-1)/k$. Show that

$$|(k\beta, n)R_n(b, z)| \leq \frac{|(\beta, s)|}{s!} (k|\beta| + k, n)|z|^n,$$

$$\left| \frac{\Gamma(\beta)R_n(b, z)}{\Gamma(k\beta)} \right| \leq \frac{|\Gamma(\beta+s)|(k|\beta| + k, n)}{s!|\Gamma(k\beta+n)|} |z|^n.$$

Note that the right side of the first inequality vanishes if $\beta = 0, -1, \dots, -s+1$.

6.2-11 Let $n \in \mathbb{N}$, $z \in \mathbb{C}^k$, and $b = (\beta, \beta, \dots, \beta) \in \mathbb{C}^k$. Show that

$$\limsup_{n \rightarrow \infty} \left| \frac{\Gamma(\beta)R_n(b, z)}{\Gamma(k\beta)} \right|^{1/n} \leq |z|.$$

6.2-12 Let $n \in \mathbb{N}$, $z \in \mathbb{C}^k$, $b = (\beta, \beta, \dots, \beta) \in \mathbb{C}^k$, and $r \in \mathbb{R}$. If $|\beta| \leq r \leq (n-k-1)/k$, show that

$$\left| \frac{\Gamma(\beta)R_n(b, z)}{\Gamma(k\beta)} \right| \leq e^{\pi k r/2} \left(\frac{n-1}{k} + 1, r \right) \frac{(kr+k, n)}{\Gamma(n-kr)} |z|^n.$$

6.2-13 Let $b, z \in \mathbb{C}^k$ and $c = \sum_{i=1}^k b_i$. Show that

$$c(c+1)R_2(b, z) = \sum_{i=1}^k b_i z_i^2 + \left(\sum_{i=1}^k b_i z_i \right)^2,$$

$$c^2(c+1)R_2(b, z) = (c+1) \left(\sum_{i=1}^k b_i z_i \right)^2 + \frac{1}{2} \sum_{i=1}^k \sum_{j=1}^k b_i b_j (z_i - z_j)^2.$$

6.2-14 Let $s, t, c \in \mathbb{C}$ and $w, z \in \mathbb{C}^k$. Assume $\sum_{i=1}^k w_i = 1$ and $cw, z^s \in \mathbb{C}_{>^k}$, where $z^s = (z_1^s, \dots, z_k^s)$. Define $\bar{z} = \sum_{i=1}^k w_i z_i$ and assume $\bar{z} \neq 0$. Show that

$$\sum_{i=1}^k w_i (z_i - \bar{z})^2 = \frac{1}{2} \sum_{i=1}^k \sum_{j=1}^k w_i w_j (z_i - z_j)^2.$$

As $|1 - z/\bar{z}| \rightarrow 0$, show that

$$R_t(cw, z^s)/(\bar{z})^{st} = 1 + \frac{1}{2} st \left(s \frac{c+t}{c+1} - 1 \right) \sum_{i=1}^k w_i (1 - z_i/\bar{z})^2 + O(|1 - z/\bar{z}|^3).$$

This result is useful when the components of z are nearly equal, as in Example 6.10-5 and Section 9.6.

6.3-1 Let $\alpha, s, t, x, y \in \mathbb{C}$ and $(\beta, \beta') \in U_2$. Show that

$$(s-x)^{-\alpha} (s-y)^{\alpha-t} R_t(\beta, \beta'; s-x, s-y)$$

is holomorphic on the s plane cut along the line segment joining x and y .

6.3-2 Let H be an open half-plane such that $0 \notin H$. Let $x, y \in H$ and $n \in \mathbb{N}$. Show that

$$R_{-1}(1, -n; x, y)/\Gamma(1-n) = n!(y-x)^n x^{-n-1}.$$

6.3-3 Prove (5.5-16) in case (ii), in which f is holomorphic.

6.3-4 Let Ω be an open circular disk in $\mathbb{C}$, and let f be holomorphic on Ω . Let $z \in \Omega^k$, $c \in U$, $w \in \mathbb{C}^k$, and $\sum_{i=1}^k w_i = 1$. Show that

$$\lim_{c \rightarrow 0} F(cw, z) = \sum_{i=1}^k w_i f(z_i).$$

6.3-5 Let $\alpha, \beta, \beta', \gamma, x, y \in \mathbb{C}$ and assume $|x| < 1$, $|y| < 1$, and $\gamma \in U$. Appell's double hypergeometric series F_1 is defined by [cf. (2.4-25)]

$$F_1(\alpha, \beta, \beta'; \gamma; x, y) = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \frac{(\alpha, m+n)(\beta, m)(\beta', n)}{(\gamma, m+n)m!n!} x^m y^n.$$

Show that

$$F_1(\alpha, \beta, \beta'; \gamma; x, y) = R_{-\alpha}(\beta, \beta', \gamma - \beta - \beta'; 1-x, 1-y, 1).$$

Deduce a transformation of F_1 which reduces to (5.9-14) if $\beta' = 0$.

6.3-6 Let $n \in \mathbb{N}$, $b = (\beta, \beta, \dots, \beta) \in \mathbb{C}^k$, and $s \in \mathbb{C}$. Let $z \in \mathbb{C}^k$ and define $\mathbb{C}' = \mathbb{C} - \text{con}(z)$. Prove that $\Gamma(\beta)R_{-n-1}(b, s-z)/\Gamma(k\beta)$ is holomorphic in (β, s) on $\mathbb{C} \times \mathbb{C}'$.

6.4-1 Let $n \in \mathbb{N}$ and $x, y, \beta, \gamma \in \mathbb{C}$. Assume $(\gamma, n) \neq 0$. Show that

$${}_2F_1(-n, \beta; \gamma; xy) = \sum_{m=0}^n \binom{n}{m} x^m (1-x)^{n-m} {}_2F_1(-m, \beta; \gamma; y).$$

6.4-2 Show that (6.4-3) and (6.4-4) are equivalent.

6.5-1 Prove (6.5-3).

6.5-2 Let $n \in \mathbb{N}$ and $x, y, \beta, \beta' \in \mathbb{C}$. Define $c = \beta + \beta'$ and assume that $x, y, x-y$, and (c, n) are nonzero. Show that

$$\begin{aligned} R_n(\beta, \beta'; x, y) &= y^n {}_2F_1(-n, \beta; c; 1-x/y) \\ &= x^n {}_2F_1(-n, \beta'; c; 1-y/x) \\ &= \frac{(\beta', n)}{(c, n)} y^n {}_2F_1(-n, \beta; 1-\beta'-n; x/y) \\ &= \frac{(\beta, n)}{(c, n)} x^n {}_2F_1(-n, \beta'; 1-\beta-n; y/x) \\ &= \frac{(\beta', n)}{(c, n)} (y-x)^n {}_2F_1\left(-n, 1-c-n; 1-\beta'-n; \frac{x}{x-y}\right) \\ &= \frac{(\beta, n)}{(c, n)} (x-y)^n {}_2F_1\left(-n, 1-c-n; 1-\beta-n; \frac{y}{y-x}\right). \end{aligned}$$

6.5-3 Let $n \in \mathbb{N}$ and $x, \beta, c \in \mathbb{C}$. Assume $x \neq 0$ and $(c, n) \neq 0$. Show that

$$\begin{aligned} {}_2F_1(-n, \beta; c; x) &= \frac{(\beta, n)}{(c, n)} (-x)^n {}_2F_1(-n, 1-c-n; 1-\beta-n; 1/x) \\ &= \frac{(c-\beta, n)}{(c, n)} {}_2F_1(-n, \beta; 1-c+\beta-n; 1-x). \end{aligned}$$

Deduce three more ways of writing this polynomial as a ${}_2F_1$ polynomial with argument $1/(1-x)$, $(x-1)/x$, or $x/(x-1)$.

6.6-1 Let $n \in \mathbb{N}$ and $b, z \in \mathbb{C}^k$, and define $c = \sum_{i=1}^k b_i$. Let γ be a counterclockwise circular path with center 0 and radius less than $|z|^{-1}$. Show that

$$(c, n)R_n(b, z) = \frac{n!}{2\pi i} \int_{\gamma} t^{-n-1} \prod_{i=1}^k (1-tz_i)^{-b_i} dt,$$

where $|\operatorname{ph}(1 - tz_i)| < \pi/2$ for $i = 1, \dots, k$. This representation will be used in Section 7.4.

6.6-2 Let $m - 1 \in \mathbb{N}$ and $x \in \mathbb{C}$. Assume $m|x| < 1$. Show that

$$(1 - x)^{-1}(1 - 2x)^{-1} \cdots (1 - mx)^{-1} = \sum_{n=m}^{\infty} \mathfrak{S}_n^{(m)} x^{n-m}.$$

6.6-3 If $m \in \mathbb{N}$ and $x \in \mathbb{C}$, show that

$$(e^x - 1)^m = m! \sum_{n=m}^{\infty} \mathfrak{S}_n^{(m)} \frac{x^n}{n!}.$$

6.6-4 Let $a, b, c, t, v \in \mathbb{C}$ and assume $c \neq 0$, $|at| < |b + (b^2 - ac)^{1/2}|$, and $|at| < |b - (b^2 - ac)^{1/2}|$. Show that

$$(at^2 + 2bt + c)^{-v} = \sum_{n=0}^{\infty} \frac{t^n (2v, n)}{c^{v+n} n!} R_n[v, v; -b + (b^2 - ac)^{1/2}, -b - (b^2 - ac)^{1/2}].$$

6.6-5 If the electrical network shown in Fig. 6.6-1 contains $2m$ identical unit resistors, find the input resistance at the terminals A, B .

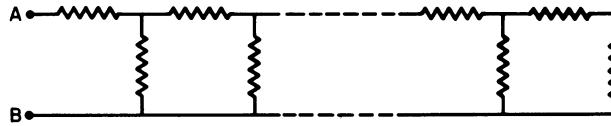


Figure 6.6-1

6.6-6 Let $n, k - 1, \kappa - 1 \in \mathbb{N}$. Let $b, z \in \mathbb{C}^k$ and $\beta, \zeta \in \mathbb{C}^\kappa$, and define $c = \sum_{i=1}^k b_i$ and $\gamma = \sum_{j=1}^\kappa \beta_j$. Let b, β stand for the array $b_1, \dots, \beta_\kappa$, and similarly for z, ζ . Show that

$$\frac{(c + \gamma, n)}{n!} R_n(b, \beta; z, \zeta) = \sum_{m=0}^n \frac{(c, m)(\gamma, n - m)}{m!(n - m)!} R_m(b, z) R_{n-m}(\beta, \zeta).$$

6.6-7 Let $n \in \mathbb{N}$ and $b, \beta, z \in \mathbb{C}^k$. Define $c = \sum_{i=1}^k b_i$ and $\gamma = \sum_{i=1}^k \beta_i$. Prove the following theorem for addition of parameters:

$$\frac{(c + \gamma, n)}{n!} R_n(b + \beta, z) = \sum_{m=0}^n \frac{(c, m)(\gamma, n - m)}{m!(n - m)!} R_m(b, z) R_{n-m}(\beta, z).$$

6.6-8 Let $n \in \mathbb{N}$ and $b, z \in \mathbb{C}^k$. Define $c = \sum_{i=1}^k b_i$ and $z^2 = (z_1^2, \dots, z_k^2)$. Show that

$$\frac{(c, n)}{n!} R_n(b, z^2) = \sum_{m=0}^{2n} \frac{(c, m)(c, 2n - m)}{m!(2n - m)!} (-1)^m R_m(b, z) R_{2n-m}(b, z).$$

6.6-9 Let $n \in \mathbb{N}$ and $b, z \in \mathbb{C}^k$. Assume $\sum_{i=1}^k b_i = n + 1$. Let γ be a positively oriented rectifiable Jordan curve such that $\operatorname{con}(z) \subset I(\gamma)$, and let f be holomorphic on $I(\gamma)$ and continuous on $\bar{I}(\gamma)$. Show that

$$F^{(n)}(b, z) = \frac{n!}{2\pi i} \int_{\gamma} f(s) \prod_{i=1}^k (s - z_i)^{-b_i} ds.$$

6.6-10 Let $\beta, \gamma, t, x \in \mathbb{C}$ and assume $|t| < 1$ and $|t - tx| < 1$. Prove the generating relation

$$(1 - t)^{\beta - \gamma} (1 - t + tx)^{-\beta} = \sum_{n=0}^{\infty} t^n \frac{(\gamma, n)}{n!} {}_2F_1(-n, \beta; \gamma; x).$$

6.6-11 If $z \in \mathbb{C}^k$, show that the elementary and complete symmetric functions of z satisfy

$$\sum_{m=0}^n (-1)^m E_m(z) C_{n-m}(z) = \delta_{n0}, \quad n = 0, 1, \dots, k,$$

where δ is the Kronecker delta. (These equations may be solved by Cramer's rule to express $C_p(z)$, $p \leq k$, as a $p \times p$ Toeplitz determinant whose nonzero entries are elementary symmetric functions multiplied by ± 1 .)

6.6-12 Let $n \in \mathbb{N}$ and $b, z \in \mathbb{C}^k$. Define $c = \sum_{i=1}^k b_i$ and e_i by (5.6-1). Show that

$$\frac{(c+1, n)}{n!} R_n(b + e_i, z) = \sum_{m=0}^n z_i^{n-m} \frac{(c, m)}{m!} R_m(b, z), \quad i = 1, 2, \dots, k.$$

6.6-13 Let $n-1 \in \mathbb{N}$ and $b, z \in \mathbb{C}^k$. Define $c = \sum_{i=1}^k b_i$. Prove the relation (due to M. D. Tobey)

$$\frac{(c, n)}{(n-1)!} R_n(b, z) = \sum_{m=0}^{n-1} \frac{(c, m)}{m!} R_m(b, z) \sum_{i=1}^k b_i z_i^{n-m}.$$

6.6-14 Let $n-1 \in \mathbb{N}$ and $z \in \mathbb{C}^k$, and define the power sums of z by $\sigma_n(z) = \sum_{i=1}^k z_i^n$. Show that the elementary and complete symmetric functions of z satisfy

$$nE_n = \sum_{m=0}^{n-1} (-1)^{n-1-m} \sigma_{n-m} E_m, \quad n = 1, 2, \dots, k,$$

$$nC_n = \sum_{m=0}^{n-1} \sigma_{n-m} C_m.$$

(The first set of equations, known as Newton's formulas, can be solved to obtain $\sigma_1, \dots, \sigma_k$ in terms of $E_1, \dots, E_k$.)

6.6-15 Let $n \in \mathbb{N}$ and $b, z \in \mathbb{C}^k$. Assume $\sum_{i=1}^k b_i = n+2$ and $0 \notin \text{con}(z)$. Let γ be a positively oriented rectifiable Jordan curve which encircles $\text{con}(z)$ but not 0. Show that

$$R_{-n-1}(b, z) = (-1)^n \frac{n+1}{2\pi i} \int_{\gamma} \log(s) \prod_{i=1}^k (s - z_i)^{-b_i} ds.$$

6.6-16 Let $x, y, z \in \mathbb{C}_{>}$. Show that

$$R_{-3/2}(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}; x, y, z) = (xyz)^{-1/2}.$$

The condition on x, y, z will be relaxed in Exercise 6.8-7.

6.6-17 Let $m \in \mathbb{Z}$, $n \in \mathbb{N}$, and $x \in \mathbb{C}_{>}$. Show that

$$\frac{2^{-n}(2n)! e^{-2m\pi x}}{x(x^2+1) \cdots (x^2+n^2)} = \int_{2m\pi}^{\infty} e^{-xt} (1 - \cos t)^n dt.$$

6.7-1 Show that $P_n(1) = 1$ and $P_n(-x) = (-1)^n P_n(x)$. More generally, show that $C_n^v(1) = (2v, n)/n!$ and $C_n^v(-x) = (-1)^n C_n^v(x)$.

6.7-2 Let $n \in \mathbb{N}$ and $v, \theta \in \mathbb{C}$. Show that

$$|P_n(\cos \theta)| \leq e^{n|\text{Im } \theta|}, \quad |C_n^v(\cos \theta)| \leq \frac{(2|v|, n)}{n!} e^{n|\text{Im } \theta|}.$$

6.7-3 The polynomials $U_n(x) = C_n^1(x)$, $n \in \mathbb{N}$, are called Chebyshev polynomials of the second kind. Show that

$$U_n(\cos \theta) = (n+1)R_n(1, 1; e^{i\theta}, e^{-i\theta}) = \frac{\sin(n+1)\theta}{\sin \theta}.$$

6.7-4 Let $t, \theta \in \mathbb{C}$ and assume $|t| < e^{-|\operatorname{Im} \theta|}$. Show that

$$-\frac{1}{2} \log(1 - 2t \cos \theta + t^2) = \sum_{n=1}^{\infty} \frac{t^n}{n} T_n(\cos \theta).$$

6.7-5 Let $n \in \mathbb{N}$ and $x, y, v \in \mathbb{C}$. Assume $(2v, N) \neq 0$. Prove the Christoffel–Darboux formula,

$$2(x-y) \sum_{n=0}^N \frac{(n+v)}{C_n^v(1)} C_n^v(x) C_n^v(y) = \frac{(N+1)}{C_N^v(1)} [C_{N+1}^v(x) C_N^v(y) - C_N^v(x) C_{N+1}^v(y)].$$

6.7-6 Let $\theta \in \mathbb{C}$ and let γ be a positively oriented contour which encircles $e^{i\theta}$ and $e^{-i\theta}$. If $n \in \mathbb{N}$ and $2v$ is a positive integer, show that

$$C_n^v(\cos \theta) = (2\pi i)^{-1} \int_{\gamma} s^{n+2v-1} (1 - 2s \cos \theta + s^2)^{-v} ds.$$

6.7-7 Show that

$$\sum_{m=0}^n P_m(\cos \theta) P_{n-m}(\cos \theta) = \frac{\sin(n+1)\theta}{\sin \theta}.$$

6.7-8 Show that

$$C_n^v(\cos 2\theta) = \sum_{m=0}^{2n} (-1)^m C_m^v(\cos \theta) C_{2n-m}^v(\cos \theta).$$

6.7-9 Let $\theta, \varphi, v \in \mathbb{C}$ and assume $\sin \theta \sin \varphi \neq 0$. Show that Gegenbauer polynomials with unrelated arguments are connected by the equation (due to Rainville and Carlitz)

$$C_n^v(\cos \theta) = \left(\frac{\sin \theta}{\sin \varphi} \right)^n \sum_{m=0}^n \frac{(2v, n)}{(2v, m)(n-m)!} \left[\frac{\sin(\varphi - \theta)}{\sin \theta} \right]^{n-m} C_m^v(\cos \varphi).$$

6.7-10 Show that Legendre polynomials satisfy

$$P_n(\cos \theta) = (2 \cos \theta)^{-n} \sum_{m=0}^n \binom{n}{m} P_m(\cos 2\theta), \quad \cos \theta \neq 0,$$

$$(\sin \theta)^n P_n(\sin \theta) = \sum_{m=0}^n \binom{n}{m} (-\cos \theta)^m P_m(\cos \theta).$$

6.7-11 Let $n-1 \in \mathbb{N}$ and $v, \theta \in \mathbb{C}$. Show that Gegenbauer polynomials satisfy the relation (due to M. D. Tobey)

$$nC_n^v(\cos \theta) = 2v \sum_{m=0}^{n-1} C_m^v(\cos \theta) \cos(n-m)\theta.$$

6.7-12 Let $\mathbf{r}$ and $\mathbf{r}'$ be vectors in $\mathbb{R}^m$, $m \geq 1$, let r and r' denote their lengths, and let θ be the angle between them ($\cos \theta = \mathbf{r} \cdot \mathbf{r}' / rr'$). Assume $r \neq r'$ and define $r_> = \max\{r, r'\}$, $r_< = \min\{r, r'\}$. If $v \in \mathbb{C}$ show that

$$|\mathbf{r} - \mathbf{r}'|^{-2v} = \sum_{n=0}^{\infty} \frac{r_<^n}{r_>^{n+2v}} C_n^v(\cos \theta).$$

(See Exercise 7.7-15.)

6.7-13 Let $n \in \mathbb{N}$, $x \in \mathbb{C}$, and $v + \frac{1}{2} \in U$. Show that Bessel functions are connected with Legendre and Gegenbauer polynomials by the relations

$$J_0(x) = \lim_{n \rightarrow \infty} P_n\left(\cos \frac{x}{n}\right), \quad \Gamma\left(\frac{1}{2} + v\right)(x/2)^{1/2-v} J_{v-1/2}(x) = \lim_{n \rightarrow \infty} C_n^v\left(\cos \frac{x}{n}\right)/C_n^v(1).$$

6.8-1 Let $t, v \in \mathbb{C}$. Show that the Legendre and Gegenbauer functions satisfy

$$P_t = P_{-t-1}, \quad [\sin \pi(t + 2v)]C_t^v = (-\sin \pi t)C_{-t-2v}^v.$$

6.8-2 Let $x, v \in \mathbb{C}_>$ and $t + 2v \in U$. Define

$$A_t^v = \frac{2^{1-2v}\Gamma(t+2v)}{[\Gamma(v)]^2\Gamma(t+1)}.$$

Show that the Gegenbauer function has the representations

$$\begin{aligned} C_t^v(x) &= A_t^v \int_0^\pi [x + (x^2 - 1)^{1/2} \cos \varphi]^t (\sin \varphi)^{2v-1} d\varphi \\ &= A_t^v \int_0^\pi [x + (x^2 - 1)^{1/2} \cos \varphi]^{-t-2v} (\sin \varphi)^{2v-1} d\varphi. \end{aligned}$$

In the case $v = \frac{1}{2}$, these integrals are known as Laplace's first and second integrals for the Legendre function $P_t = C_t^{1/2}$. In this case $A_t^{1/2} = 1/\pi$ and t is unrestricted. Whenever the quantity in brackets is raised to a power that is a nonnegative integer, x is unrestricted.

6.8-3 Let $v \in \mathbb{C}_>$ and $t + 2v \in U$. Define A_t^v as in Exercise 6.8-2. If $0 < \theta < \pi/2$, or if $0 < \theta < \pi$ and $t \in \mathbb{N}$, show that

$$C_t^v(\cos \theta) = A_t^v 2^v (\sin \theta)^{1-2v} \int_0^\theta \cos[(t+v)\varphi] (\cos \varphi - \cos \theta)^{v-1} d\varphi.$$

In the case $v = \frac{1}{2}$ this is called the Mehler-Dirichlet integral for $P_t = C_t^{1/2}$, and t is unrestricted.

6.8-4 Let $\alpha, v, t, x \in \mathbb{C}$, set $x = \cos \theta$, and assume $|t| < e^{-|\operatorname{Im} \theta|}$. Show that

$$\frac{\Gamma(\alpha+1)}{\Gamma(\alpha+2v)} (1-2tx+t^2)^{v/2} C_\alpha^v \left[\frac{1-tx}{(1-2tx+t^2)^{1/2}} \right] = \sum_{n=0}^{\infty} t^n \frac{(-\alpha, n)}{\Gamma(2v+n)} C_n^v(x).$$

6.8-5 Let $t, \theta, x \in \mathbb{C}$ and assume $|\operatorname{Re} \theta| < \pi$. Show that

$$\begin{aligned} \frac{\sin t\theta}{t \sin \theta} &= R_{t-1}(1, 1; e^{i\theta}, e^{-i\theta}), & t \sin \theta \neq 0, \\ \cos(\pi t/2) &= (1+t)R_t(1, 1; i, -i), \\ \frac{\theta}{\sin \theta} &= R_{-1}(1, 1; e^{i\theta}, e^{-i\theta}), \\ \log x &= (x-1)R_{-1}(1, 1; x, 1), & x \in \mathbb{C}_0, \\ \arctan x &= xR_{-1}(1, 1; 1+ix, 1-ix), & 1 \pm ix \in \mathbb{C}_0, \\ \operatorname{arccoth} x &= R_{-1}(1, 1; x+1, x-1), & x-1 \in \mathbb{C}_0. \end{aligned}$$

Compare Exercises 6.9-16 and 6.9-17.

6.8-6 Let $x, y \in \mathbb{C}_>$. Show that

$$R_{-1}(1, 1; x^2, y^2) = R_{-1}\left(1, 1; x \frac{x+y}{2}, y \frac{x+y}{2}\right).$$

6.8-7 Let $x, y, z \in \mathbb{C}_0$. Show that

$$R_{-3/2}(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}; x, y, z) = (xyz)^{-1/2},$$

$$R_{-5/2}(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}; x, y, z) = \tfrac{1}{3}(xyz)^{-3/2}(xy + yz + zx).$$

6.8-8 Let $a, a' \in \mathbb{C}_>$, $b \in U_k$, $z \in \mathbb{C}_0^k$, and assume $a + a' = \sum_{i=1}^k b_i$. Show that

$$B(a, a') R_{-a}(b, z) = \int_0^\infty t^{a-1} \prod_{i=1}^k (1 + tz_i)^{-b_i} dt.$$

6.9-1 Show that $C_{2n}^v(0) = (-1)^n (v, n)/n!$ and $C_{2n+1}^v(0) = 0$. In particular, $P_{2n}(0) = (-1)^n (\tfrac{1}{2}, n)/n!$.

6.9-2 If $n \in \mathbb{N}$ and $\beta \in \mathbb{C}$, show that

- (1) $R_{2n+1}(\beta, -2\beta - 2n; 2, 1) = 0, \quad (\beta, 2n+1) \neq 0,$
- (2) $R_{2n}(\beta, 1 - 2\beta - 2n; 2, 1) = \frac{(2n)! (\beta, n)}{n! (\beta, 2n)}, \quad (\beta, 2n) \neq 0,$
- (3) ${}_2F_1(-2n, \beta; 2\beta; 2) = (\tfrac{1}{2}, n)/(\beta + \tfrac{1}{2}, n), \quad (\beta + \tfrac{1}{2}, n) \neq 0,$
- (4) ${}_2F_1(-2n, \beta; 1 - \beta - 2n; -1) = \frac{(2n)! (\beta, n)}{n! (\beta, 2n)}, \quad (\beta, 2n) \neq 0,$
- (5) ${}_2F_1(-2n, 1 - 2\beta - 2n; 1 - \beta - 2n; \tfrac{1}{2}) = (\tfrac{1}{2}, n)(\beta, n)/(\beta, 2n), \quad (\beta, 2n) \neq 0.$

6.9-3 Let $m \in \mathbb{N}$ and let $[m/2]$ denote the largest integer not exceeding $m/2$. Let $\beta, x, y \in \mathbb{C}$ and assume $(\beta + \tfrac{1}{2}, [m/2]) \neq 0$. Show that

$$R_m(\beta, \beta; x, y) = \frac{m!}{(2\beta, m)} \sum_{p=0}^{[m/2]} \frac{(\beta, m-p)}{p! (m-2p)!} (-xy)^p (x+y)^{m-2p}.$$

If $y = -x$, this reduces to Theorem 6.9-1.

6.9-4 Show that the coefficients of Gegenbauer's polynomial are given by

$$C_n^v(z) = \sum_{m=0}^{[n/2]} \frac{(v, n-m)}{m! (n-2m)!} (-1)^m (2z)^{n-2m}.$$

6.9-5 Let $t, \beta, \delta, x, y \in \mathbb{C}$ and assume $2\beta + \delta \in U$ and $|x-y| < |x+y|$. Show that

$$R_t\left(\beta, \beta, \delta; x, y, \frac{x+y}{2}\right) = \left(\frac{x+y}{2}\right)^t {}_3F_2\left[\frac{-t}{2}, \frac{1-t}{2}, \beta; \beta + \frac{\delta}{2}, \beta + \frac{\delta+1}{2}; \left(\frac{x-y}{x+y}\right)^2\right].$$

If $\delta = 0$, this reduces to (6.9-7).

6.9-6 Let $n \in \mathbb{N}$, $\beta + \tfrac{1}{2} \in U$, and $x, y \in \mathbb{C}$. Show that

$$R_n(\beta, \beta; x+y, x-y) = \sum_{m=0}^{[n/2]} \binom{n}{2m} \frac{(\tfrac{1}{2}, m)}{(\beta + \tfrac{1}{2}, m)} x^{n-2m} y^{2m}.$$

6.9-7 Let $n \in \mathbb{N}$ and $x, y \in \mathbb{C}$. Let $\sigma, \beta + \frac{1}{2} \in \mathbb{C}_>$. Show that

$$\begin{aligned} & \mathbf{B}\left(\beta + \frac{1}{2}, \sigma\right) R_n(\beta + \sigma, \beta + \sigma; x + y, x - y) \\ &= \int_0^1 u^{\beta-1/2} (1-u)^{\sigma-1} R_n(\beta, \beta; x + y\sqrt{u}, x - y\sqrt{u}) du. \end{aligned}$$

6.9-8 Let $n \in \mathbb{N}$ and $\beta, x, y \in \mathbb{C}$. Show that

$$\begin{aligned} & \left(\beta + \frac{1}{2}, n\right) R_{2n}(\beta, \beta; x, y) = (\beta + n, n) R_n\left[\frac{1}{2} - \beta - n, \frac{1}{2} - n; \left(\frac{x+y}{2}\right)^2, \left(\frac{x-y}{2}\right)^2\right], \\ & \left(\beta + \frac{1}{2}, n\right) R_{2n+1}(\beta, \beta; x, y) = (1 + \beta + n, n) \frac{x+y}{2} \\ & \quad \times R_n\left[\frac{1}{2} - \beta - n, -\frac{1}{2} - n; \left(\frac{x+y}{2}\right)^2, \left(\frac{x-y}{2}\right)^2\right]. \end{aligned}$$

6.9-9 Let $n \in \mathbb{N}$ and $v, x \in \mathbb{C}$. Show that

$$\begin{aligned} C_{2n}^v(x) &= (-1)^n \frac{(v, n)}{n!} {}_2F_1\left(-n, v + n; \frac{1}{2}; x^2\right), \\ C_{2n+1}^v(x) &= (-1)^n \frac{(v, n+1)}{n!} 2x {}_2F_1\left(-n, v + n + 1; \frac{3}{2}; x^2\right). \end{aligned}$$

6.9-10 If $n \in \mathbb{N}$ and $\theta \in \mathbb{C}$, show that

$$\begin{aligned} \cos 2n\theta &= {}_2F_1(-n, n; \tfrac{1}{2}; \sin^2 \theta) = (-1)^n {}_2F_1(-n, n; \tfrac{1}{2}; \cos^2 \theta), \\ \cos(2n+1)\theta &= \cos \theta {}_2F_1(-n, n+1; \tfrac{1}{2}; \sin^2 \theta) \\ &= (-1)^n (2n+1) \cos \theta {}_2F_1(-n, n+1; \tfrac{3}{2}; \cos^2 \theta). \end{aligned}$$

6.9-11 If $n \in \mathbb{N}$ and $\theta \in \mathbb{C}$, show that

$$\begin{aligned} \sin 2n\theta &= n \sin 2\theta {}_2F_1(1-n, 1+n; \tfrac{3}{2}; \sin^2 \theta) \\ &= (-1)^{n+1} n \sin 2\theta {}_2F_1(1-n, 1+n; \tfrac{3}{2}; \cos^2 \theta), \\ \sin(2n+1)\theta &= (2n+1) \sin \theta {}_2F_1(-n, 1+n; \tfrac{3}{2}; \sin^2 \theta) \\ &= (-1)^n \sin \theta {}_2F_1(-n, 1+n; \tfrac{1}{2}; \cos^2 \theta). \end{aligned}$$

6.9-12 Show that the Fibonacci numbers are sums of binomial coefficients:

$$F_{n+1} = \sum_{m=0}^{\lfloor n/2 \rfloor} \binom{n-m}{m}, \quad L_n = \sum_{m=0}^{\lfloor n/2 \rfloor} \frac{n}{n-m} \binom{n-m}{m}.$$

6.9-13 Let $k-2 \in \mathbb{N}$ and define $\omega = e^{i2\pi/k}$. Let $\beta \in \mathbb{C}$ and $m \in \mathbb{N}$. Prove the following generalization of Theorem 6.9-1:

$$\begin{aligned} R_{km+p}(\beta, \dots, \beta; \omega, \omega^2, \dots, \omega^k) &= 0, \quad p = 1, 2, \dots, k-1, \quad (k\beta, km+p) \neq 0, \\ R_{km}(\beta, \dots, \beta; \omega, \omega^2, \dots, \omega^k) &= \frac{(km)! (\beta, m)}{m! (k\beta, km)}, \quad (k\beta, km) \neq 0. \end{aligned}$$

(The arguments of the R polynomials are the k th roots of unity.)

6.9-14 Let $z_1, \dots, z_k$ be the vertices of a regular polygon in the complex plane. Denote the center of the polygon by λ and one of the vertices by $\lambda + x$. Let f be holomorphic on an open circular disk containing the polygon. Assume $k\beta \in U$. Show that

$$F(\beta, \dots, \beta; z_1, \dots, z_k) = \sum_{m=0}^{\infty} f^{(km)}(\lambda) \frac{(\beta, m)}{(k\beta, km)} \frac{x^{km}}{m!}.$$

In particular, show that

$$S(\beta, \dots, \beta; z_1, \dots, z_k) = e^{\lambda} {}_0F_{k-1} \left[\beta + \frac{1}{k}, \dots, \beta + \frac{k-1}{k}; \left(\frac{x}{k}\right)^k \right].$$

If $|x| < |\lambda|$ and $a \in \mathbb{C}$, show that

$$\begin{aligned} R_{-a}(\beta, \dots, \beta; z_1, \dots, z_k) \\ = \lambda^{-a} {}_kF_{k-1} \left[\frac{a}{k}, \frac{a+1}{k}, \dots, \frac{a+k-1}{k}; \beta + \frac{1}{k}, \dots, \beta + \frac{k-1}{k}; \left(\frac{-x}{\lambda}\right)^k \right]. \end{aligned}$$

6.9-15 Let $\beta, \beta', x, y \in \mathbb{C}$ and assume $2\beta + 2\beta' \in U$. Show that

$$S\left(\beta, \beta, 2\beta'; x, y, \frac{x+y}{2}\right) = e^{(x+y)/2} {}_1F_2 \left[\beta; \beta + \beta', \beta + \beta' + \frac{1}{2}; \left(\frac{x-y}{4}\right)^2 \right].$$

6.9-16 Show that

$$\begin{aligned} \arccos x &= (1-x^2)^{1/2} R_C(x^2, 1), & 0 \leq x \leq 1, \\ \arctan x &= x R_C(1, 1+x^2), & -\infty < x < \infty, \\ \operatorname{arccot} x &= R_C(x^2, x^2+1), & 0 \leq x < \infty, \\ \operatorname{arsinh} x &= x R_C(1+x^2, 1), & -\infty < x < \infty, \\ \operatorname{arccosh} x &= (x^2-1)^{1/2} R_C(x^2, 1), & x \geq 1, \\ \operatorname{arctanh} x &= x R_C(1, 1-x^2), & -1 < x < 1, \\ \operatorname{arcoth} x &= R_C(x^2, x^2-1), & x > 1. \end{aligned}$$

6.9-17 Let $t, \theta, \varphi \in \mathbb{C} - \{0\}$ and assume $|\operatorname{Re} \theta| < \frac{1}{2}\pi$ and $|\operatorname{Im} \varphi| < \frac{1}{2}\pi$. Show that

$$\begin{aligned} \frac{\sin t\theta}{t \sin \theta} &= R_{(t-1)/2} \left(\frac{1}{2} + \frac{1}{2}t, 1 - \frac{1}{2}t; \cos^2 \theta, 1 \right), \\ \frac{\theta}{\sin \theta} &= R_C(\cos^2 \theta, 1), & \pi &= 2R_C(0, 1) = 4R_C(1, 2) = 6R_C(3, 4), \\ \theta &= R_C(\cot^2 \theta, \csc^2 \theta), & \varphi &= R_C(\coth^2 \varphi, \operatorname{csch}^2 \varphi). \end{aligned}$$

6.9-18 Let $x, y \in \mathbb{C}_{>}$ and $\varepsilon = |1 - x/y|$. As $\varepsilon \rightarrow 0$, show that

$$\left(\frac{x+2y}{3}\right) R_C(x^2, y^2) = 1 + \frac{1}{5} \left(\frac{x-y}{x+2y}\right)^2 + O(\varepsilon^3).$$

6.9-19 Prove that ${}_0F_1(c; x)/\Gamma(c)$ is entire in c and x .

6.9-20 Let $n \in \mathbb{N}$ and $x, c, \mu \in \mathbb{C}$. Prove (6.9-23) and deduce that

$$\left(\frac{1}{x} \frac{d}{dx}\right)^n [x^{-\mu} J_{\mu}(x)] = (-1)^n x^{-\mu-n} J_{\mu+n}(x).$$

If $x \in \mathbb{C}_0$, show that

$$\frac{d^n}{dx^n} \left[x^{c-1} \frac{{}_0F_1(c; x)}{\Gamma(c)} \right] = x^{c-n-1} \frac{{}_0F_1(c-n; x)}{\Gamma(c-n)},$$

and deduce that

$$\left(\frac{1}{x} \frac{d}{dx}\right)^n [x^{\mu} J_{\mu}(x)] = x^{\mu-n} J_{\mu-n}(x).$$

6.9-21 Let $x \in \mathbb{C}$ and $n \in \mathbb{Z}$. Prove (6.9-24) and (6.9-25).

6.10-1 Let $\alpha, \beta, z \in \mathbb{C}$ and assume $\alpha + \beta + \frac{1}{2} \in U$ and $\operatorname{Re} z < \frac{1}{2}$. Show that

$${}_2F_1(2\alpha, 2\beta; \alpha + \beta + \frac{1}{2}; z) = {}_2F_1[\alpha, \beta; \alpha + \beta + \frac{1}{2}; 4z(1-z)].$$

6.10-2 Compute $R_K(1, 2) = 0.834627 \dots$

6.10-3 Show that

$$[\Gamma(\frac{1}{4})]^2 = (2\pi)^{3/2} R_K(1, 2).$$

Compute $\Gamma(\frac{1}{4}) = 3.62561 \dots$

6.10-4 Show that the period of a simple pendulum is $T = T_s R_K(\cos^2 \frac{1}{2} \varphi, 1)$, where T_s is the period of small oscillations and $\varphi < \pi$ is the amplitude of oscillation. If the pendulum is released from rest in a horizontal position ($\varphi = \frac{1}{2}\pi$), find T/T_s to five decimal places.

6.10-5 Let $k \in \mathbb{C}$ and $k' = (1 - k^2)^{1/2}$, and assume $k' \in \mathbb{C}_>$. Show that the first complete elliptic integral satisfies $K(k) = (1/k') K(ik/k')$. Prove also the Landen transformation,

$$K(k) = \frac{2}{1+k'} K\left(\frac{1-k'}{1+k'}\right).$$

6.10-6 Let $a, b, c, t, v \in \mathbb{C}$ and assume $c \neq 0$ and $|at| < |b \pm (b^2 - ac)^{1/2}|$. Show that

$$(at^2 + 2bt + c)^{-v} = \sum_{n=0}^{\infty} \frac{t^n 2^n (v, n)}{c^{v+n} n!} R_n \left[\frac{1}{2} - v - n, \frac{1}{2} - v - n; -b + (ac)^{1/2}, -b - (ac)^{1/2} \right].$$

6.10-7 Let $n \in \mathbb{N}$ and $x, v \in \mathbb{C}$, and assume $(v + \frac{1}{2}, n) \neq 0$. Show that

$$C_n^v(x) = \frac{(-2)^{-n} (2v, n)}{(v + \frac{1}{2}, n)} (x^2 - 1)^{n/2} C_n^{1/2-v-n}[x(x^2 - 1)^{-1/2}].$$

6.10-8 Let $n \in \mathbb{N}$ and $\alpha, x, y \in \mathbb{C}$. Show that

$$\begin{aligned} & (\tfrac{1}{2} + \alpha, n) R_n(-\alpha - n, -\alpha - n; x^2, y^2) \\ &= 2^{-2n} (1 + 2\alpha, n) R_n[\tfrac{1}{2} + \alpha, \tfrac{1}{2} + \alpha; (x+y)^2, (x-y)^2]. \end{aligned}$$

6.10-9 Let $n \in \mathbb{N}$. Show that the Fibonacci and Lucas numbers are given by

$$F_{n+1} = R_n(-\tfrac{1}{2} - n, -\tfrac{1}{2} - n; 1 + 2i, 1 - 2i), \quad L_n = R_n(\tfrac{1}{2} - n, \tfrac{1}{2} - n; 1 + 2i, 1 - 2i).$$

6.10-10 Let $n \in \mathbb{N}$ and $x, v \in \mathbb{C}$, and assume $(v + \frac{1}{2}, n) \neq 0$. Show that

$$C_n^v(x) = \frac{(2v, n)}{n!} {}_2F_1\left(-n, 2v + n; \frac{1}{2} + v; \frac{1-x}{2}\right).$$

6.10-11 Let $n \in \mathbb{N}$ and $\theta \in \mathbb{C}$. Assume $\sin \theta \neq 0$. Show that

$$2^n R_n\left(-\frac{1}{2} - n, -\frac{1}{2} - n; \cos \theta + 1, \cos \theta - 1\right) = \frac{\sin(n+1)\theta}{\sin \theta}.$$

6.10-12 Let $\beta + \frac{1}{2} \in U$ and $x, y \in \mathbb{C}_>$. Show that

$$R_{1/2-\beta}[\beta, \beta; x^2, y^2] = \left(\frac{x+y}{2}\right)^{1-2\beta}, \quad R_{-1/2-\beta}[\beta, \beta; x^2, y^2] = (xy)^{-1} \left(\frac{x+y}{2}\right)^{1-2\beta}$$

6.10-13 Compute $\pi/4 = R_C(1, 2) = 0.785398 \dots$ by using (6.10-29).

6.11-1 A charge Q is distributed uniformly on a circle in $\mathbb{R}^3$, every point of which has radial coordinate a and polar angle α . The charge creates an electric field with potential

$$V(r, \theta) = \frac{Q}{2\pi} \int_0^{2\pi} [a^2 + r^2 - 2ar(\cos \alpha \cos \theta + \sin \alpha \sin \theta \cos \beta)]^{-1/2} d\beta.$$

Let $r_{\pm}^2 = a^2 + r^2 - 2ar \cos(\theta \pm \alpha)$ denote the square of the maximum (+) or minimum (−) distance to the circle from the point where V is evaluated. Show that

$$V = QR_K(r_+^2, r_-^2) = Q/\rho,$$

where ρ is the arithmetic-geometric mean of r_+ and r_- . If $r \neq a$, $r_> = \max\{r, a\}$, and $r_< = \min\{r, a\}$, show that

$$V(r, \theta) = Q \sum_{n=0}^{\infty} \frac{r_<^n}{r_>^{n+1}} P_n(\cos \alpha) P_n(\cos \theta).$$